

# Chapter 41

## Animal Nutrition

PowerPoint® Lecture Presentations for

# Biology

*Eighth Edition*

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# Overview: The Need to Feed

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- Food is taken in, taken apart, and taken up in the process of animal **nutrition**.
- In general, animals fall into three categories:
  - **Herbivores** eat mainly autotrophs (plants and algae).
  - **Carnivores** eat other animals.
  - **Omnivores** regularly consume animals as well as plants or algal matter.

# **An animal's diet must supply chemical energy, organic molecules, and essential nutrients**

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- An animal's diet provides **chemical energy**, which is converted into **ATP** and powers processes in the body.
- Animals need a source of **organic carbon** and **organic nitrogen** in order to construct ***organic molecules***.
- **Essential nutrients** are required by cells and must be **obtained from dietary sources**.

# Essential Nutrients

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- There are four classes of **essential nutrients**:
  - Essential amino acids
  - Essential fatty acids
  - Vitamins
  - Minerals

# *Essential Amino Acids*

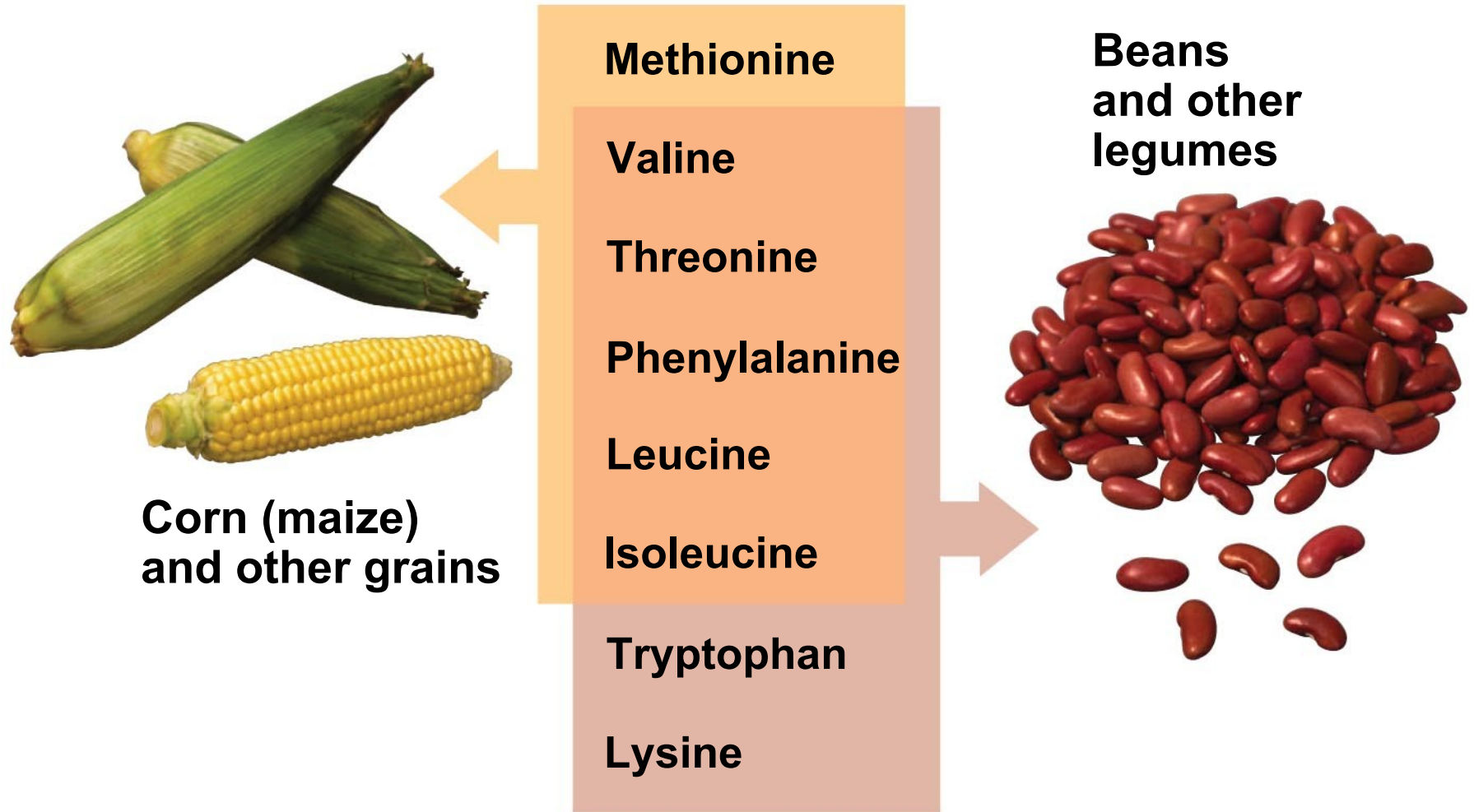
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- Animals require 20 amino acids and can synthesize about half from molecules in their diet.
- The remaining amino acids, the **essential amino acids** must be obtained from food in preassembled form.
- A diet that provides insufficient essential amino acids causes malnutrition called protein deficiency.

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- Meat, eggs, and cheese provide all the essential amino acids and are thus “complete” proteins.
  - Most plant proteins are incomplete in amino acid makeup.
  - Individuals who eat only plant proteins need to eat specific plant combinations to get all essential amino acids.
  - Some animals have adaptations that help them through periods when their bodies demand extraordinary amounts of protein.

# Essential amino acids from a vegetarian diet

## 8 Essential amino acids for adults



# *Essential Fatty Acids*

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- Animals can synthesize most of the fatty acids they need.
- The **essential fatty acids** are certain unsaturated fatty acids that must be obtained from the diet.
- Deficiencies in fatty acids are rare.



# *Vitamins*

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- **Vitamins** are **organic molecules** required in the diet in small amounts. Many vitamins function as **coenzymes**.
- 13 vitamins essential to humans have been identified.
- Vitamins are grouped into two categories: fat-soluble and water-soluble.

**Table 41.1 Vitamin Requirements of Humans**

| Vitamin                                | Major Dietary Sources                                                                                  | Major Functions in the Body                                                                                                   | Symptoms of Deficiency or Extreme Excess                                                                                              |
|----------------------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Water-Soluble Vitamins</b>          |                                                                                                        |                                                                                                                               |                                                                                                                                       |
| Vitamin B <sub>1</sub> (thiamine)      | Pork, legumes, peanuts, whole grains                                                                   | Coenzyme used in removing CO <sub>2</sub> from organic compounds                                                              | Beriberi (nerve disorders, emaciation, anemia)                                                                                        |
| Vitamin B <sub>2</sub> (riboflavin)    | Dairy products, meats, enriched grains, vegetables                                                     | Component of coenzymes FAD and FMN                                                                                            | Skin lesions such as cracks at corners of mouth                                                                                       |
| Niacin (B <sub>3</sub> )               | Nuts, meats, grains                                                                                    | Component of coenzymes NAD <sup>+</sup> and NADP <sup>+</sup>                                                                 | Skin and gastrointestinal lesions, nervous disorders <b>Liver damage</b>                                                              |
| Vitamin B <sub>6</sub> (pyridoxine)    | Meats, vegetables, whole grains                                                                        | Coenzyme used in amino acid metabolism                                                                                        | Irritability, convulsions, muscular twitching, anemia <b>Unstable gait, numb feet, poor coordination</b>                              |
| Pantothenic acid (B <sub>5</sub> )     | Most foods: meats, dairy products, whole grains, etc.                                                  | Component of coenzyme A                                                                                                       | Fatigue, numbness, tingling of hands and feet                                                                                         |
| Folic acid (folacin) (B <sub>9</sub> ) | Green vegetables, oranges, nuts, legumes, whole grains                                                 | Coenzyme in nucleic acid and amino acid metabolism                                                                            | Anemia, birth defects <b>May mask deficiency of vitamin B<sub>12</sub></b>                                                            |
| Vitamin B <sub>12</sub>                | Meats, eggs, dairy products                                                                            | Coenzyme in nucleic acid metabolism; maturation of red blood cells                                                            | Anemia, nervous system disorders                                                                                                      |
| Biotin                                 | Legumes, other vegetables, meats                                                                       | Coenzyme in synthesis of fat, glycogen, and amino acids                                                                       | Scaly skin inflammation, neuromuscular disorders                                                                                      |
| Vitamin C (ascorbic acid)              | Fruits and vegetables, especially citrus fruits, broccoli, cabbage, tomatoes, green peppers            | Used in collagen synthesis (such as for bone, cartilage, gums); antioxidant; aids in detoxification; improves iron absorption | Scurvy (degeneration of skin, teeth, blood vessels), weakness, delayed wound healing, impaired immunity <b>Gastrointestinal upset</b> |
| <b>Fat-Soluble Vitamins</b>            |                                                                                                        |                                                                                                                               |                                                                                                                                       |
| Vitamin A (retinol)                    | Provitamin A (beta-carotene) in deep green and orange vegetables and fruits; retinal in dairy products | Component of visual pigments; maintenance of epithelial tissues; antioxidant; helps prevent damage to cell membranes          | Blindness and increased death rate <b>Headache, irritability, vomiting, hair loss, blurred vision, liver and bone damage</b>          |
| Vitamin D                              | Dairy products, egg yolk; also made in human skin in presence of sunlight                              | Aids in absorption and use of calcium and phosphorus; promotes bone growth                                                    | Rickets (bone deformities) in children, bone softening in adults <b>Brain, cardiovascular, and kidney damage</b>                      |
| Vitamin E (tocopherol)                 | Vegetable oils, nuts, seeds                                                                            | Antioxidant; helps prevent damage to cell membranes                                                                           | Degeneration of the nervous system                                                                                                    |
| Vitamin K (phylloquinone)              | Green vegetables, tea; also made by colon bacteria                                                     | Important in blood clotting                                                                                                   | Defective blood clotting <b>Liver damage and anemia</b>                                                                               |

# *Minerals*

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- **Minerals** are simple **inorganic** nutrients, usually required in small amounts. Minerals serve a variety of important functions including **enzymes cofactors**.

**Table 41.2 Mineral Requirements of Humans**

| Mineral                              | Major Dietary Sources                                      | Major Functions in the Body                                                            | Symptoms of Deficiency*                                                        |                                                     |
|--------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------|
| Greater than 200 mg per day required | Calcium (Ca)                                               | Dairy products, dark green vegetables, legumes                                         | Bone and tooth formation, blood clotting, nerve and muscle function            | Retarded growth, possibly loss of bone mass         |
|                                      | Phosphorus (P)                                             | Dairy products, meats, grains                                                          | Bone and tooth formation, acid-base balance, nucleotide synthesis              | Weakness, loss of minerals from bone, calcium loss  |
|                                      | Sulfur (S)                                                 | Proteins from many sources                                                             | Component of certain amino acids                                               | Symptoms of protein deficiency                      |
|                                      | Potassium (K)                                              | Meats, dairy products, many fruits and vegetables, grains                              | Acid-base balance, water balance, nerve function                               | Muscular weakness, paralysis, nausea, heart failure |
|                                      | Chlorine (Cl)                                              | Table salt                                                                             | Acid-base balance, formation of gastric juice, nerve function, osmotic balance | Muscle cramps, reduced appetite                     |
|                                      | Sodium (Na)                                                | Table salt                                                                             | Acid-base balance, water balance, nerve function                               | Muscle cramps, reduced appetite                     |
|                                      | Magnesium (Mg)                                             | Whole grains, green leafy vegetables                                                   | Cofactor; ATP bioenergetics                                                    | Nervous system disturbances                         |
| Iron (Fe)                            | Meats, eggs, legumes, whole grains, green leafy vegetables | Component of hemoglobin and of electron carriers in energy metabolism; enzyme cofactor | Iron-deficiency anemia, weakness, impaired immunity                            |                                                     |
| Fluorine (F)                         | Drinking water, tea, seafood                               | Maintenance of tooth (and probably bone) structure                                     | Higher frequency of tooth decay                                                |                                                     |
| Zinc (Zn)                            | Meats, seafood, grains                                     | Component of certain digestive enzymes and other proteins                              | Growth failure, skin abnormalities, reproductive failure, impaired immunity    |                                                     |
| Copper (Cu)                          | Seafood, nuts, legumes, organ meats                        | Enzyme cofactor in iron metabolism, melanin synthesis, electron transport              | Anemia, cardiovascular abnormalities                                           |                                                     |
| Manganese (Mn)                       | Nuts, grains, vegetables, fruits, tea                      | Enzyme cofactor                                                                        | Abnormal bone and cartilage                                                    |                                                     |
| Iodine (I)                           | Seafood, dairy products, iodized salt                      | Component of thyroid hormones                                                          | Goiter (enlarged thyroid)                                                      |                                                     |
| Cobalt (Co)                          | Meats and dairy products                                   | Component of vitamin B <sub>12</sub>                                                   | None, except as B <sub>12</sub> deficiency                                     |                                                     |
| Selenium (Se)                        | Seafood, meats, whole grains                               | Enzyme cofactor; antioxidant functioning in close association with vitamin E           | Muscle pain, possibly heart muscle deterioration                               |                                                     |
| Chromium (Cr)                        | Brewer's yeast, liver, seafood, meats, some vegetables     | Involved in glucose and energy metabolism                                              | Impaired glucose metabolism                                                    |                                                     |
| Molybdenum (Mo)                      | Legumes, grains, some vegetables                           | Enzyme cofactor                                                                        | Disorder in excretion of nitrogen-containing compounds                         |                                                     |

\*All of these minerals are also harmful when consumed in excess.

# Dietary Deficiencies

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- **Undernourishment** is the result of a diet that consistently supplies less chemical energy than the body requires.
- **Malnourishment** is the long-term absence from the diet of one or more essential nutrients.

# *Undernourishment*

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- An undernourished individual will
  - Use up stored fat and carbohydrates
  - Break down its own proteins
  - Lose muscle mass
  - Suffer protein deficiency of the brain
  - Die or suffer irreversible damage.



# *Malnourishment*

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- Malnourishment can cause deformities, disease, and death. Malnourishment can be corrected by changes to a diet.
- Insights into human nutrition have come from *epidemiology, the study of human health and disease in populations.*
- Neural tube defects were found to be the result of a deficiency in folic acid in pregnant mothers.

# Question: Can diet influence the frequency of birth defects?

## RESULTS

| Group                                    | Number of infants/fetuses studied | Infants/fetuses with a neural tube defect |
|------------------------------------------|-----------------------------------|-------------------------------------------|
| Vitamin supplements (experimental group) | 141                               | 1 (0.7%)                                  |
| No vitamin supplements (control group)   | 204                               | 12 (5.9%)                                 |



# The main stages of food processing are ingestion, digestion, absorption, and elimination

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- **Ingestion** is the act of **eating**. There are a variety of types of eating:
  - Suspension feeders
  - Substrate feeders
    - Fluid feeders
    - Bulk feeders

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# Suspension Feeders

- Many aquatic animals are **suspension feeders**, which sift small food particles from the water.



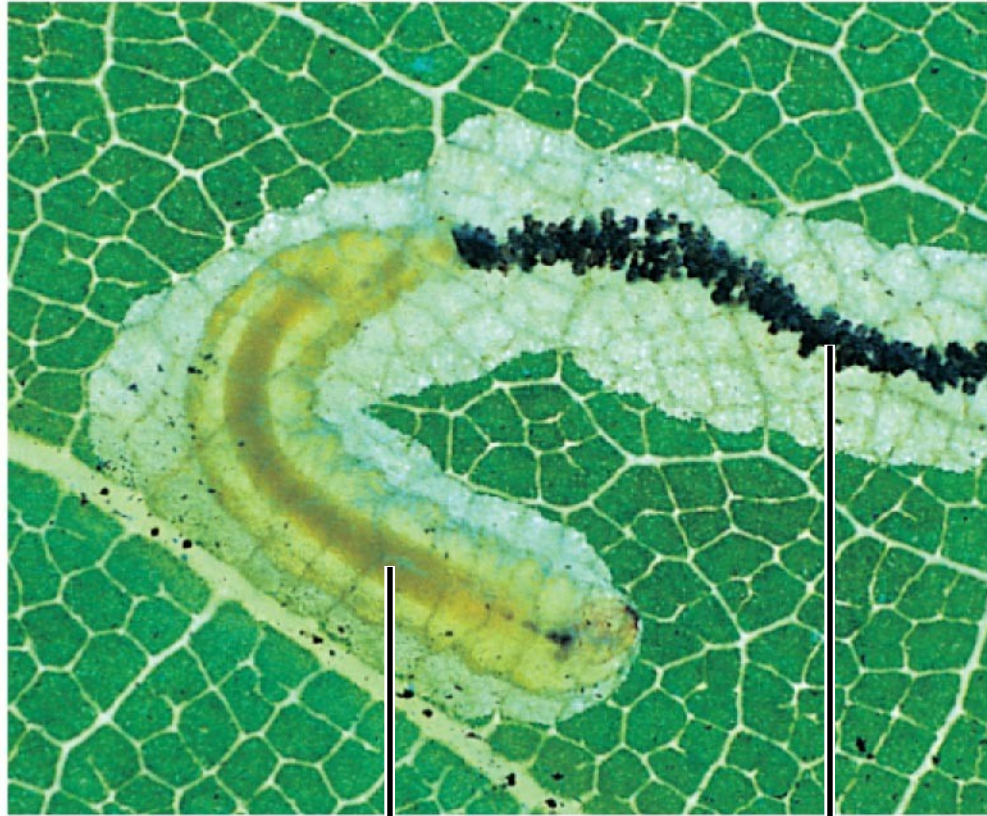
**Humpback whale, a suspension feeder**

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**Substrate feeders are animals that live in or on their food source.**

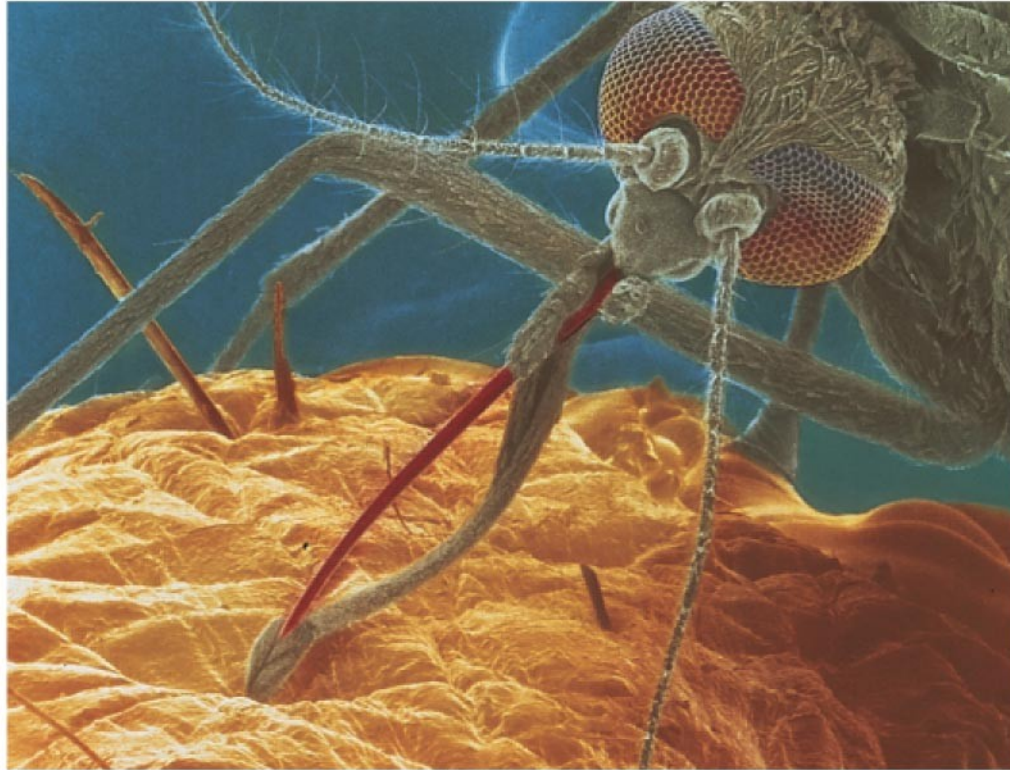
**Leaf miner caterpillar, a substrate feeder**



**Caterpillar**

**Feces**

**Fluid feeders suck nutrient-rich fluid from a living host.**



**Mosquito, a fluid feeder**

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**Bulk feeders eat relatively large pieces of food.**

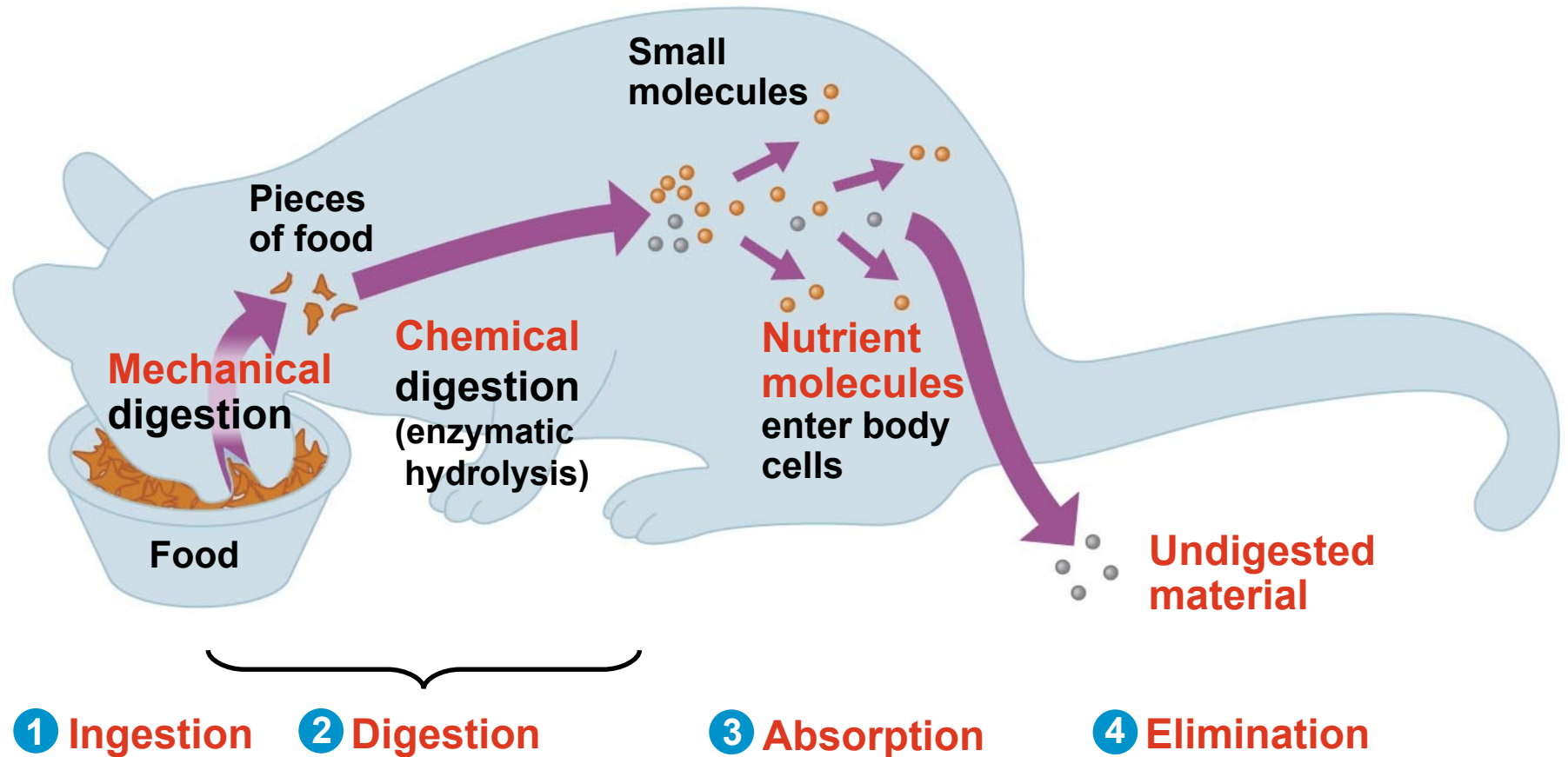


**Rock python, a bulk feeder**

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- **Digestion** is the *process of breaking food down into soluble molecules* - small enough to absorb.
    - In **chemical digestion**, the process of **enzymatic hydrolysis** splits bonds in molecules with the addition of water.
  - **Absorption** is uptake of nutrients by body cells.
  - **Elimination** is the passage of undigested material out of the digestive compartment.

# The four stages of food processing



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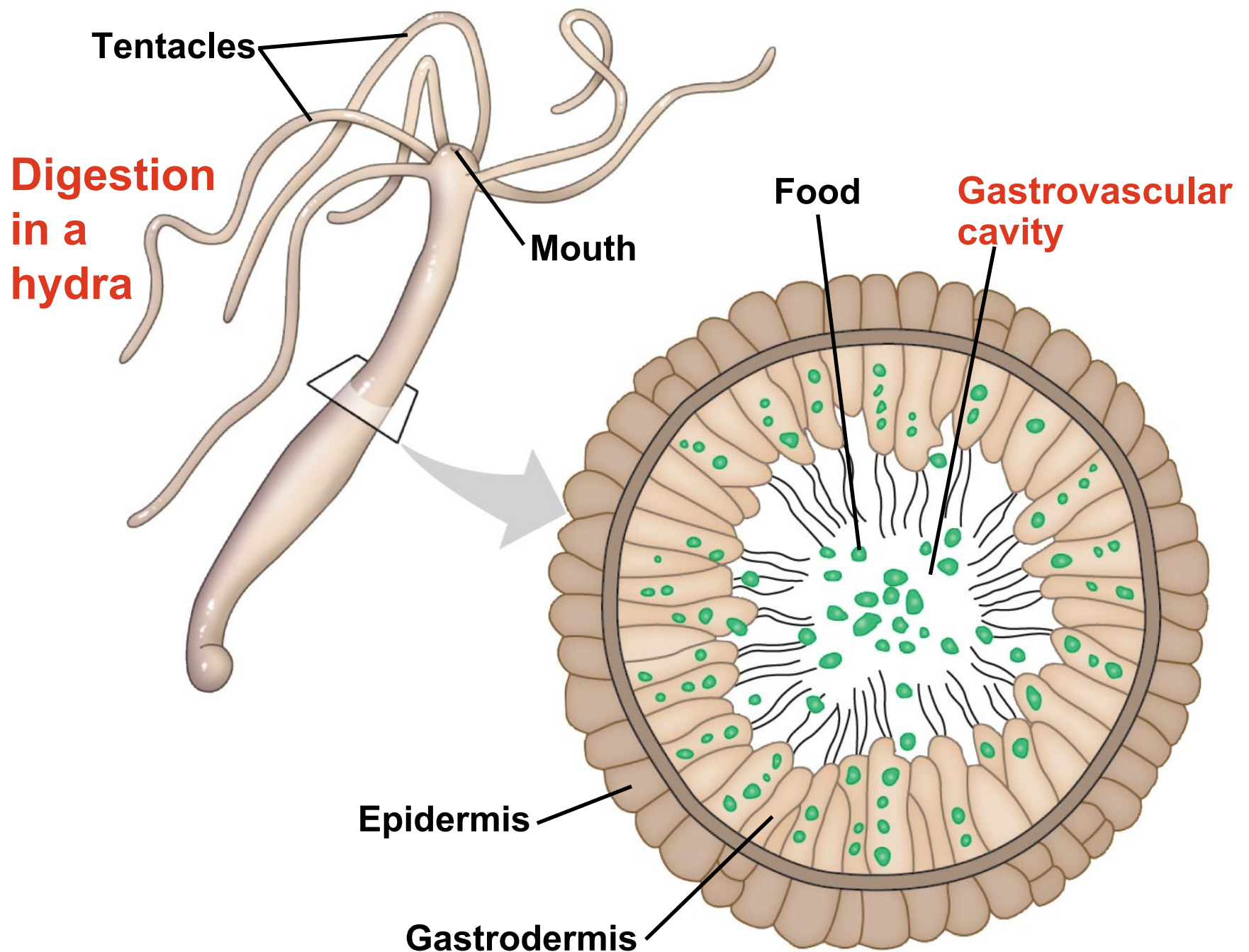
**Mechanical & Chemical  
Digestion**



# Digestive Compartments

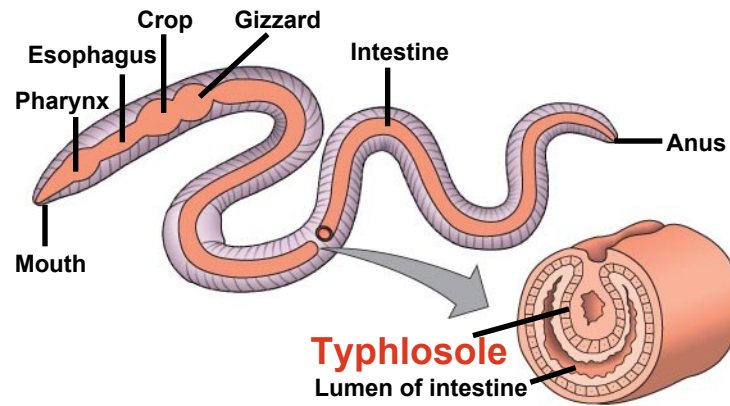
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- Most animals process food in specialized compartments. These compartments reduce the risk of an animal digesting its own cells and tissues.
- **Intracellular digestion**, food particles are engulfed by endocytosis and digested within food vacuoles.
- **Extracellular digestion** is the breakdown of food particles outside of cells. It occurs in compartments that are continuous with the outside of the animal's body.

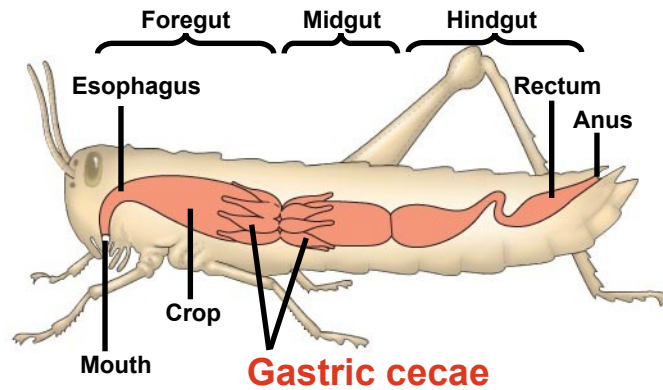


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- Animals with simple body plans have a **gastrovascular cavity** with only **one opening** that functions as **mouth / anus**. This gastrovascular cavity functions in both digestion and distribution of nutrients.
  - More complex animals have a digestive tube with **two openings**, a mouth and an anus.
  - This **one way digestive tube** is called a **complete digestive tract** or an **alimentary canal**. It can have specialized regions that carry out digestion and absorption in a stepwise, **efficient** fashion.
-

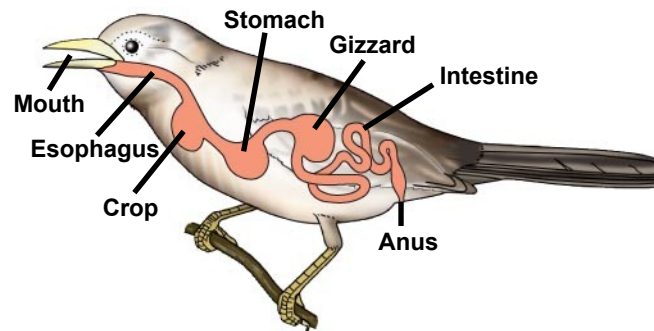
# Variation in alimentary canals



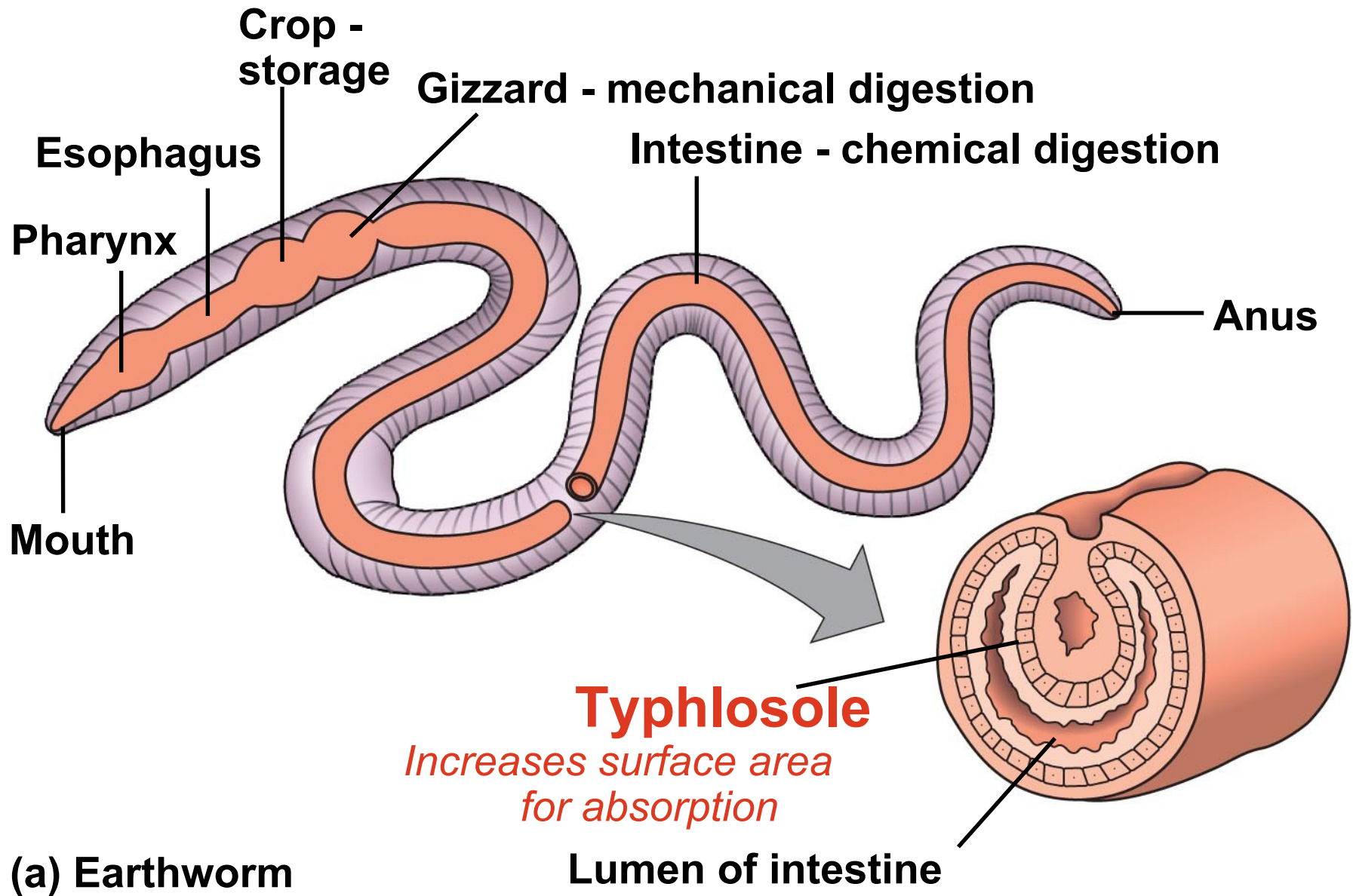
(a) Earthworm



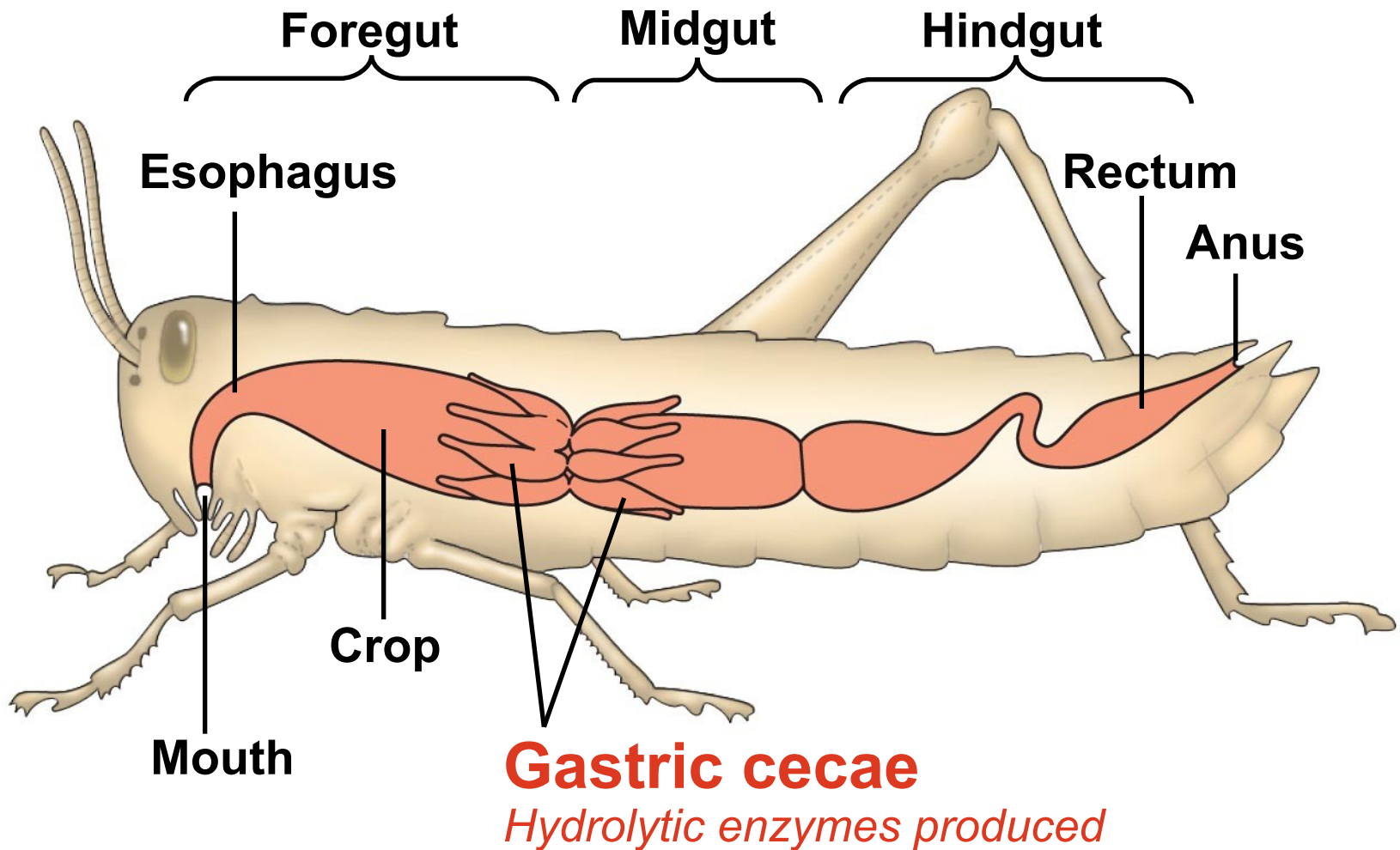
(b) Grasshopper



(c) Bird

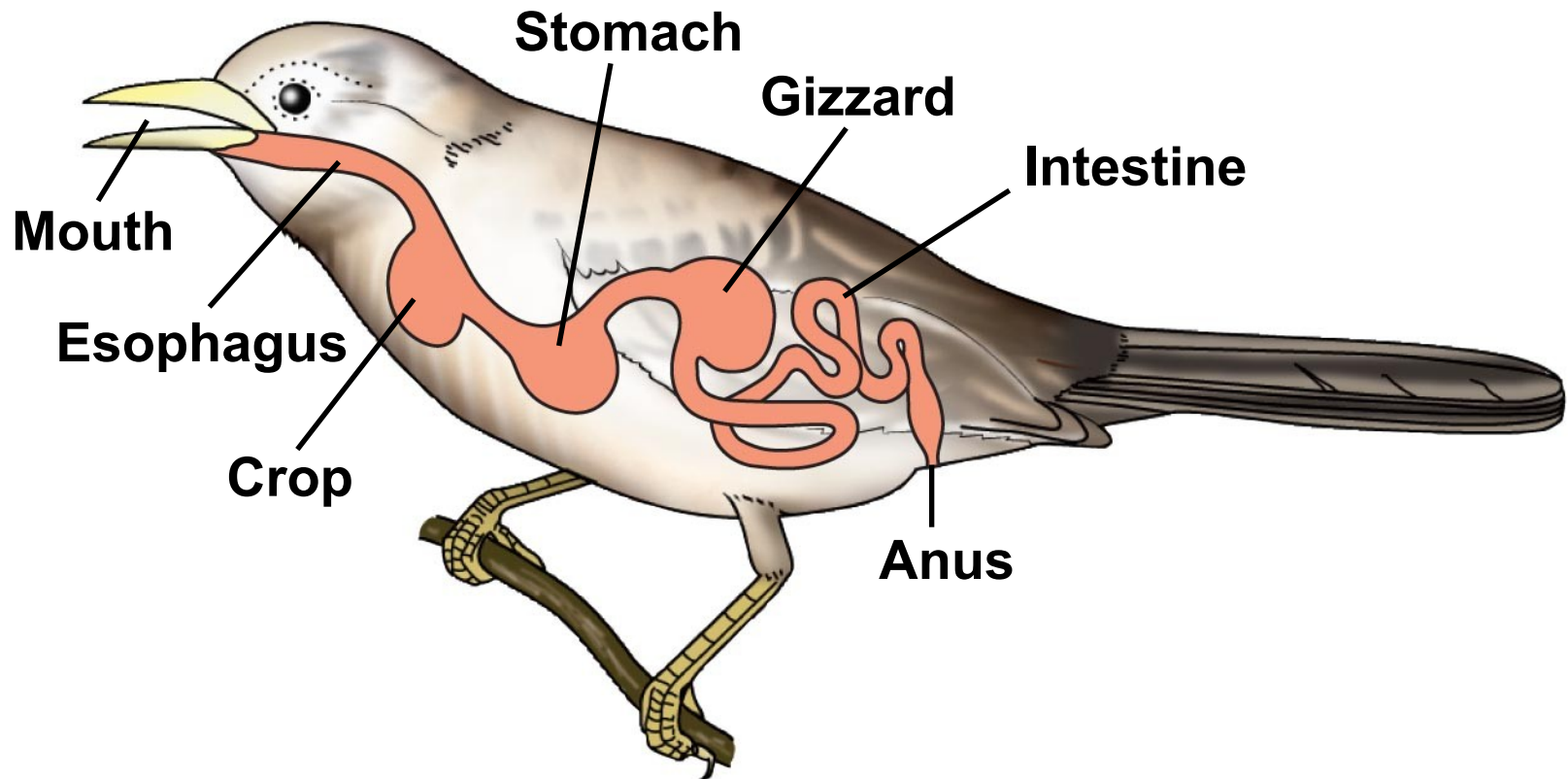






**(b) Grasshopper**





**(c) Bird**

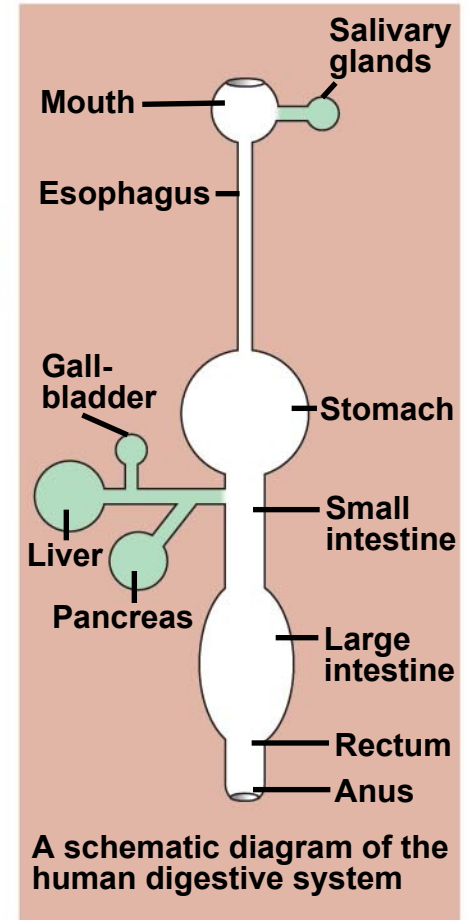
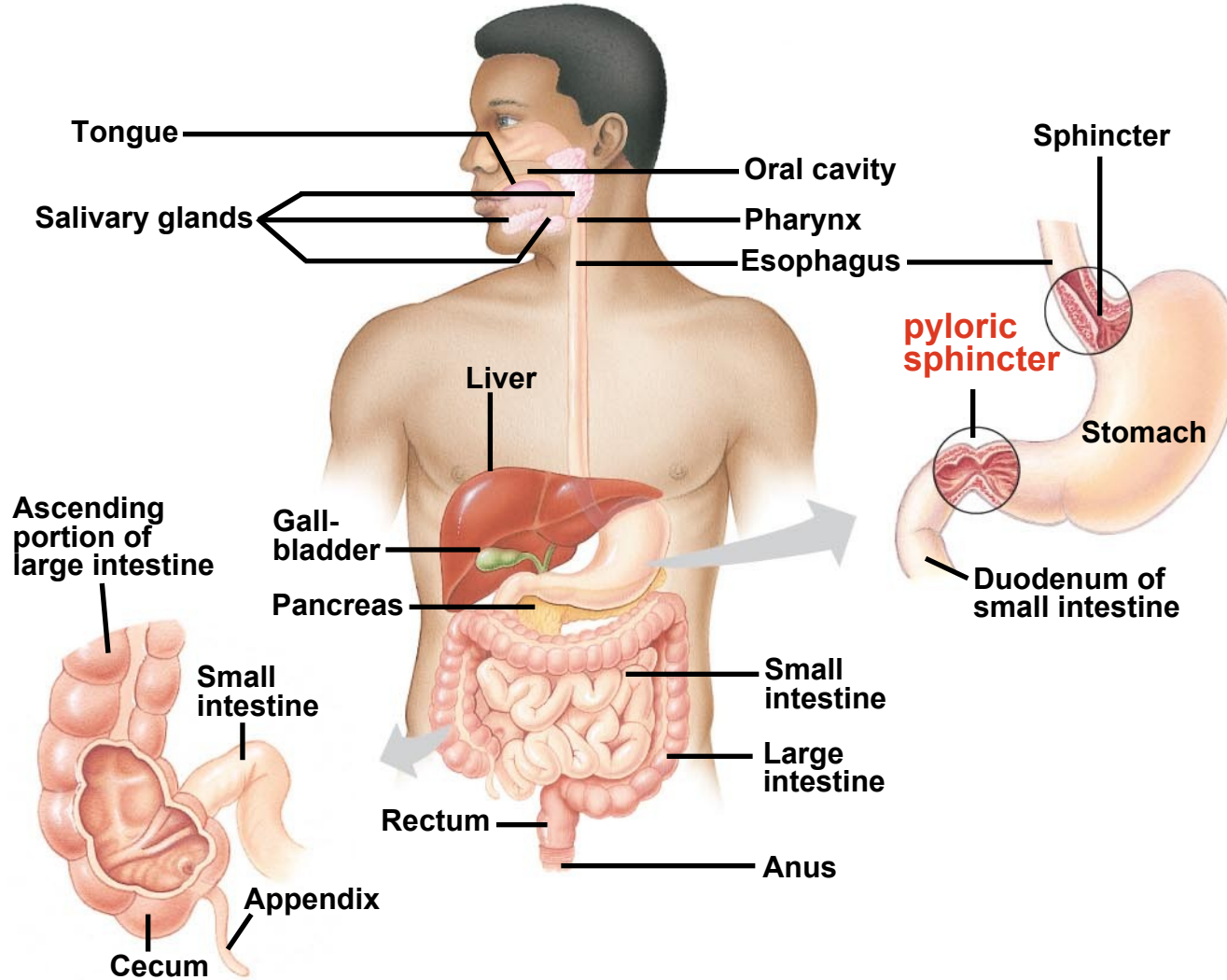
# Organs specialized for sequential stages of food processing form the mammalian digestive system

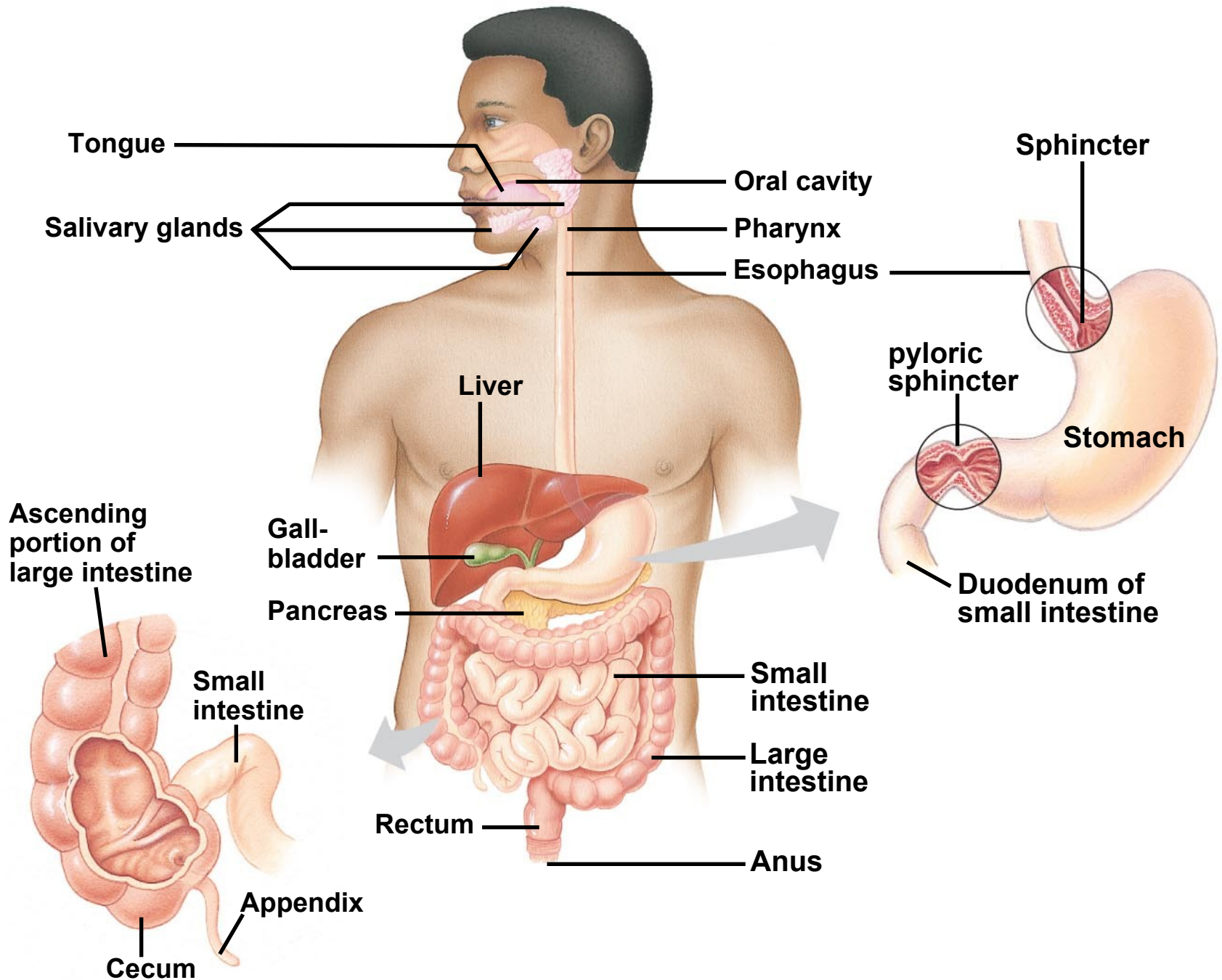
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- The mammalian digestive system consists of an **alimentary canal** and **accessory glands** that secrete digestive juices through ducts.
- Mammalian accessory glands are the **salivary glands**, the **pancreas**, the **liver**, and the **gallbladder**.

- 
- Food is pushed along by **peristalsis**, rhythmic contractions of smooth muscles in the wall of the alimentary canal.
  - **Valves** called **sphincters** regulate the movement of material between compartments.

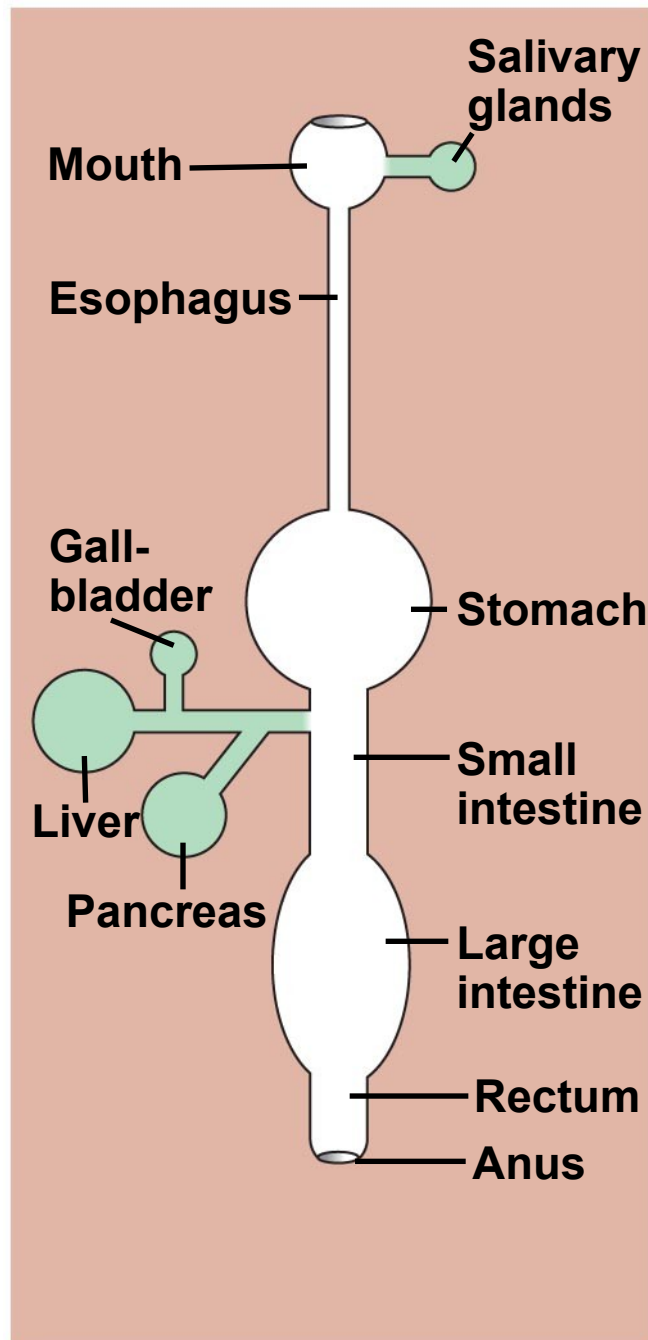
# human digestive system







# A Schematic Diagram of the Human Digestive System



# The Oral Cavity, Pharynx, and Esophagus

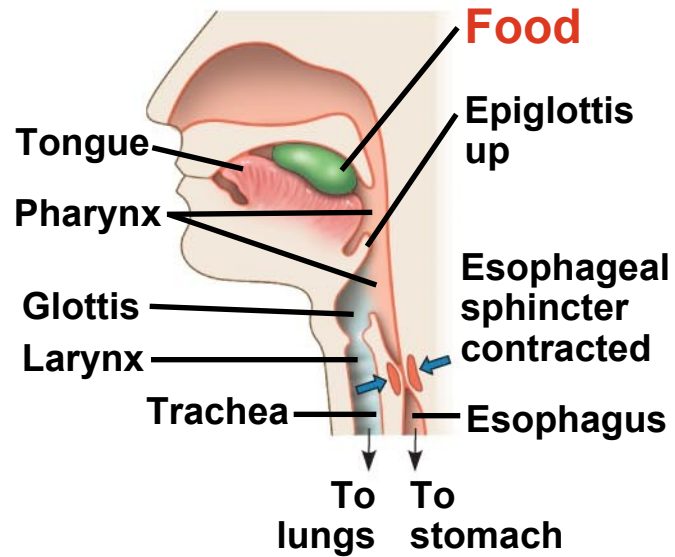
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- The first stage of digestion is mechanical and takes place in the **oral cavity**.
- **Salivary glands** deliver saliva to lubricate food.
- Teeth chew food into smaller particles. This is *mechanical digestion* that *increases the surface area exposed to the enzyme: **salivary amylase***, initiating breakdown of glucose polymers = **carbohydrate digestion**.

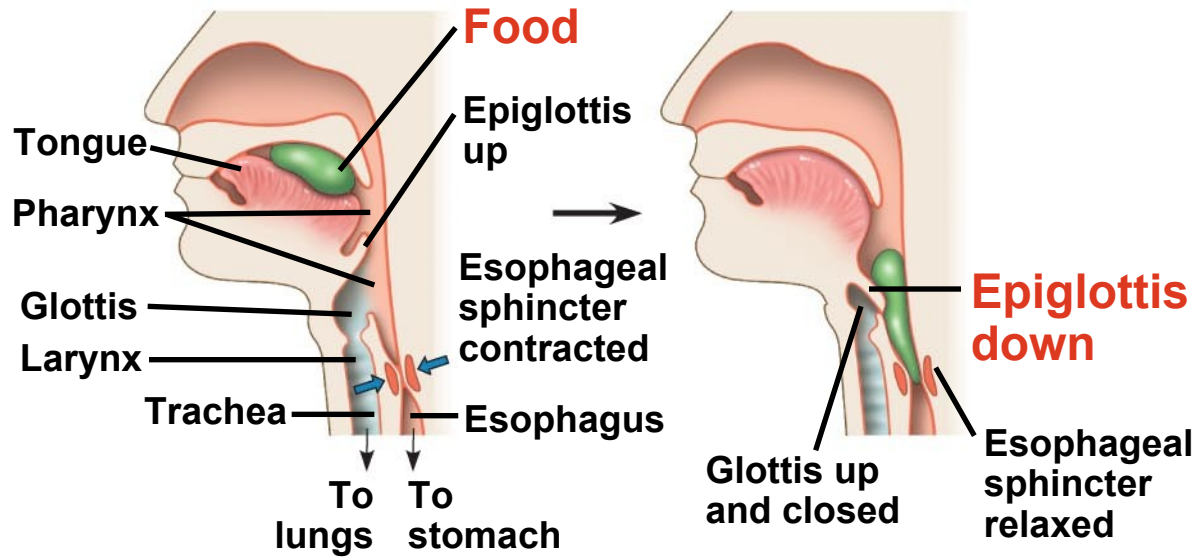
- 
- The tongue shapes food into a **bolus** and provides help with swallowing.
  - The region we call our throat is the **pharynx**, a junction that opens to both the esophagus and the trachea (windpipe).
  - The trachea leads to the lungs.

- 
- The **esophagus** conducts food from the pharynx down to the stomach by **peristalsis**.
  - *Swallowing causes the epiglottis to block entry to the trachea*, and the bolus is guided by the larynx, the upper part of the respiratory tract.
  - Coughing occurs when the swallowing reflex fails and food or liquids reach the windpipe.

# From mouth to stomach: the **swallowing reflex** and peristalsis

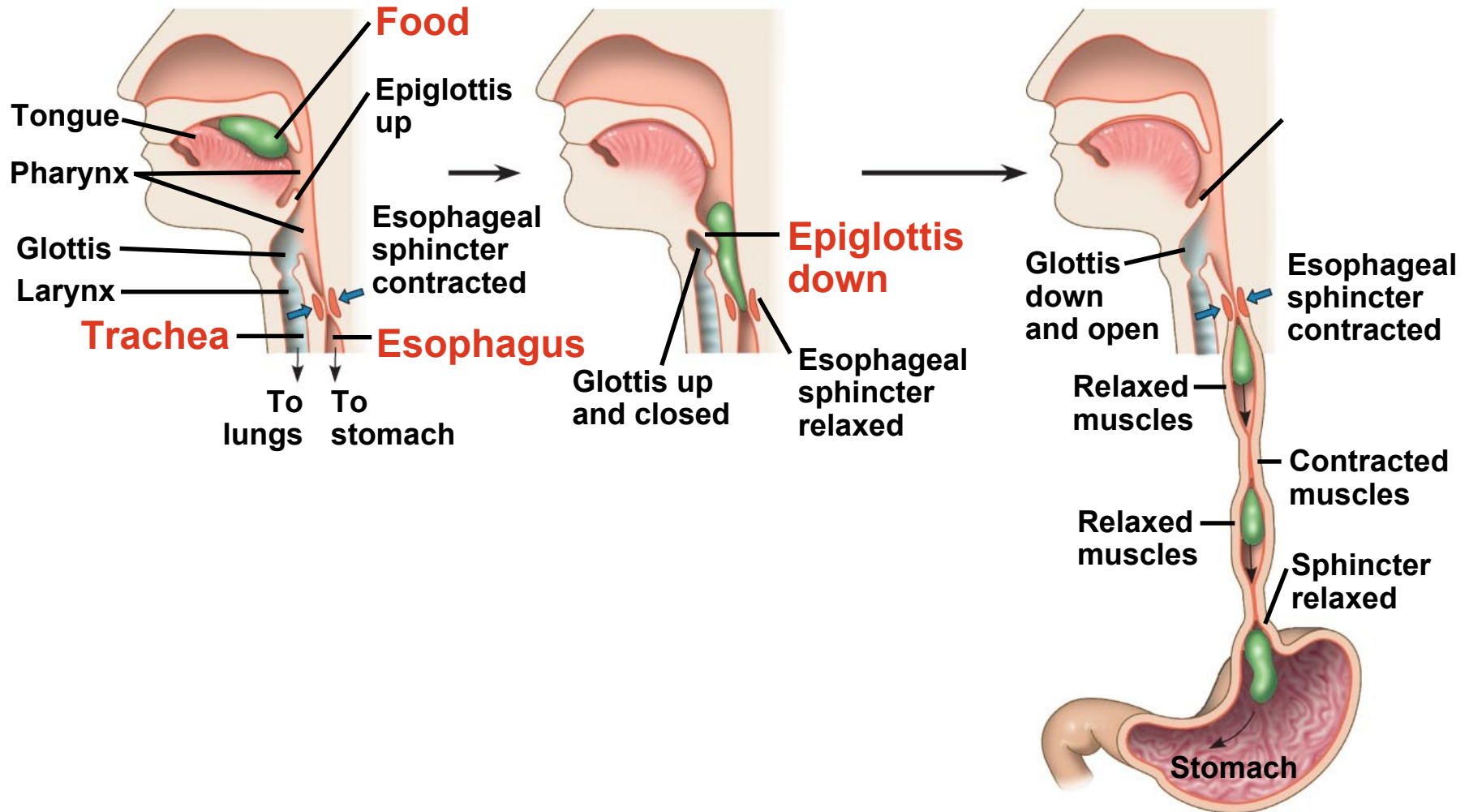


# From mouth to stomach: the **swallowing reflex** and peristalsis





# From mouth to stomach: the **swallowing reflex** and peristalsis

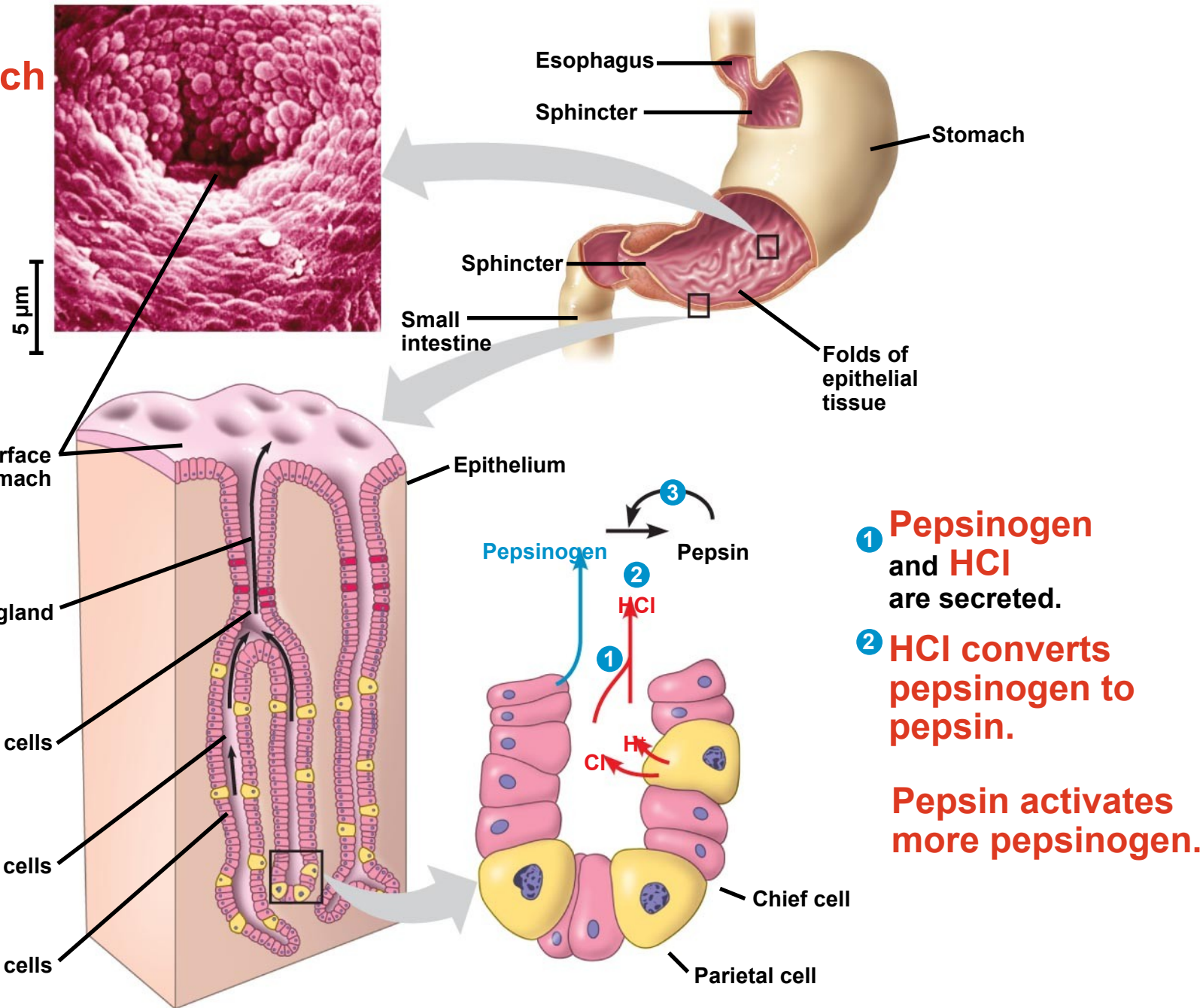


# *Chemical Digestion in the Stomach*

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- The **stomach** stores food and secretes **gastric juice**, which converts a meal to acid **chyme**.
- Gastric juice is made up of **hydrochloric acid** and the **enzyme pepsin**.
- *Parietal cells* secrete hydrogen and chloride ions separately.
- *Chief cells* secrete inactive **pepsinogen**, which is activated to pepsin when mixed with hydrochloric acid in the stomach.
- *Mucus protects the stomach lining* from gastric juice.

# The stomach and its secretions



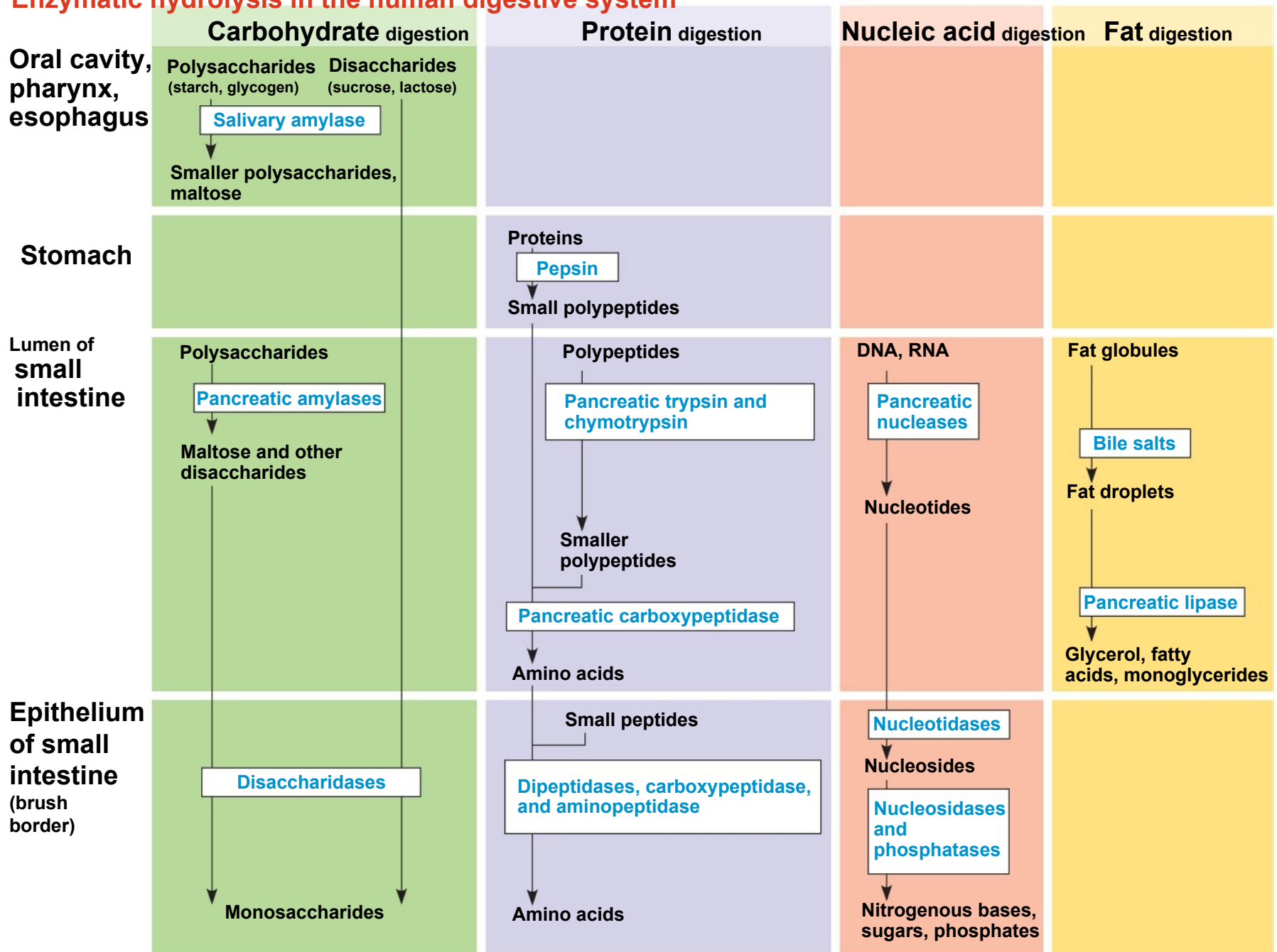
- 
- Gastric ulcers, lesions in the lining, are caused mainly by the bacterium *Helicobacter pylori*.
  - Coordinated contraction and relaxation of stomach muscle churn the stomach's contents.
  - *Sphincters* prevent chyme from entering the esophagus and *regulate* its entry into the small intestine.

# Digestion in the Small Intestine

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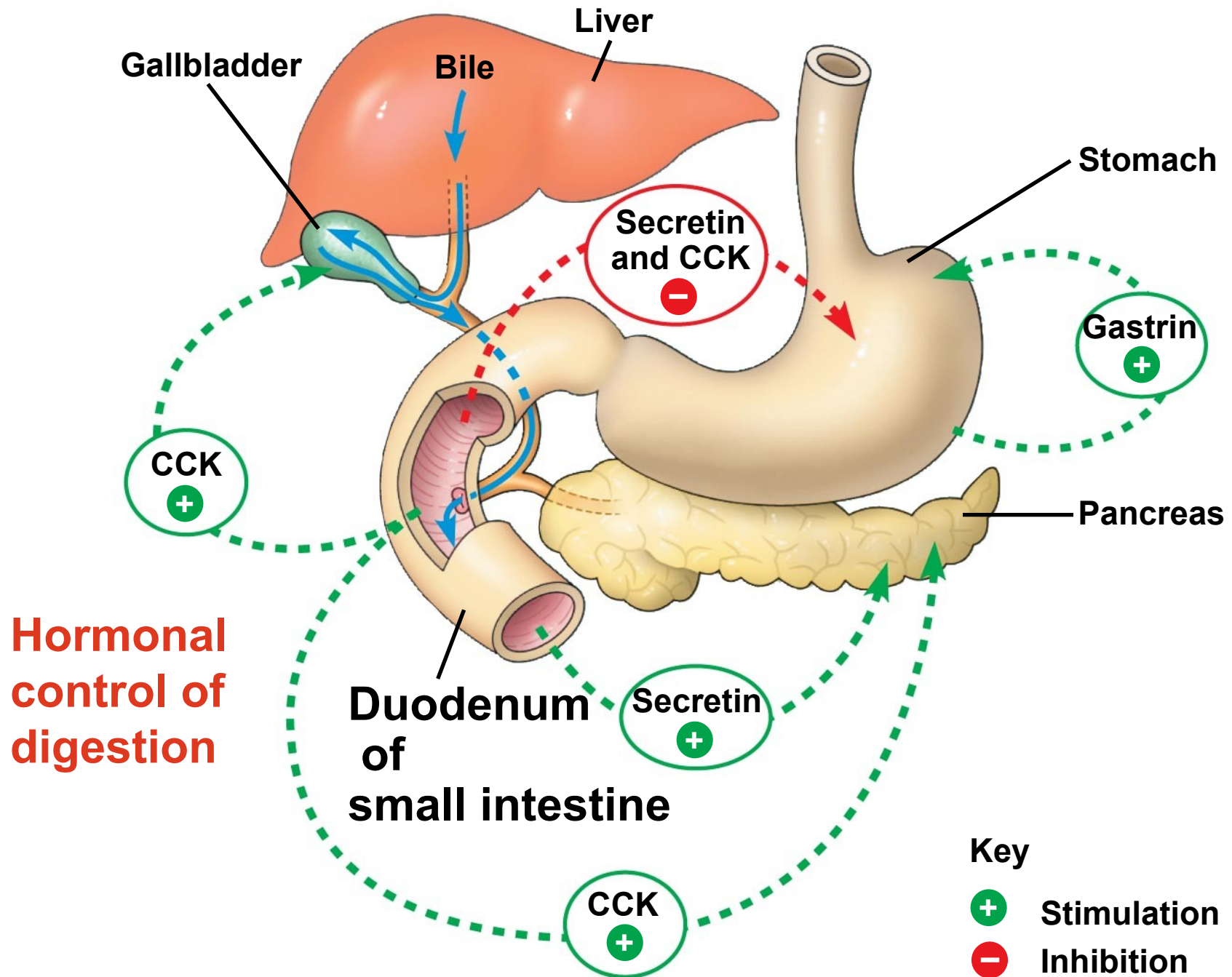
- The **small intestine** is the longest section of the alimentary canal.
- It is **the major organ of digestion and absorption.**

# Enzymatic hydrolysis in the human digestive system





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- The first portion of the small intestine is the **duodenum**, where acid chyme from the stomach mixes with digestive juices from the pancreas, liver, gallbladder, and the small intestine itself.



# *Pancreatic Secretions*

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- The **pancreas** produces proteases trypsin and chymotrypsin, protein-digesting enzymes that are activated after entering the duodenum.
- Its **solution is alkaline** and **neutralizes the acidic chyme**.

# *Bile Production by the Liver*

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- In the small intestine, *bile* aids in digestion and absorption of fats. Bile emulsifies fat. This is physical NOT chemical digestion. *Fat emulsification increases the surface area for chemical digestion of fats by lipases.*
- Bile is made in the **liver** and stored in the **gallbladder**.

# *Secretions of the Small Intestine*

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- The epithelial lining of the duodenum, called the brush border, produces several digestive enzymes.
- *Enzymatic digestion is completed* as peristalsis moves the chyme and digestive juices along the small intestine.
- *Most digestion occurs in the duodenum*; the jejunum and ileum function mainly in absorption of nutrients and water.

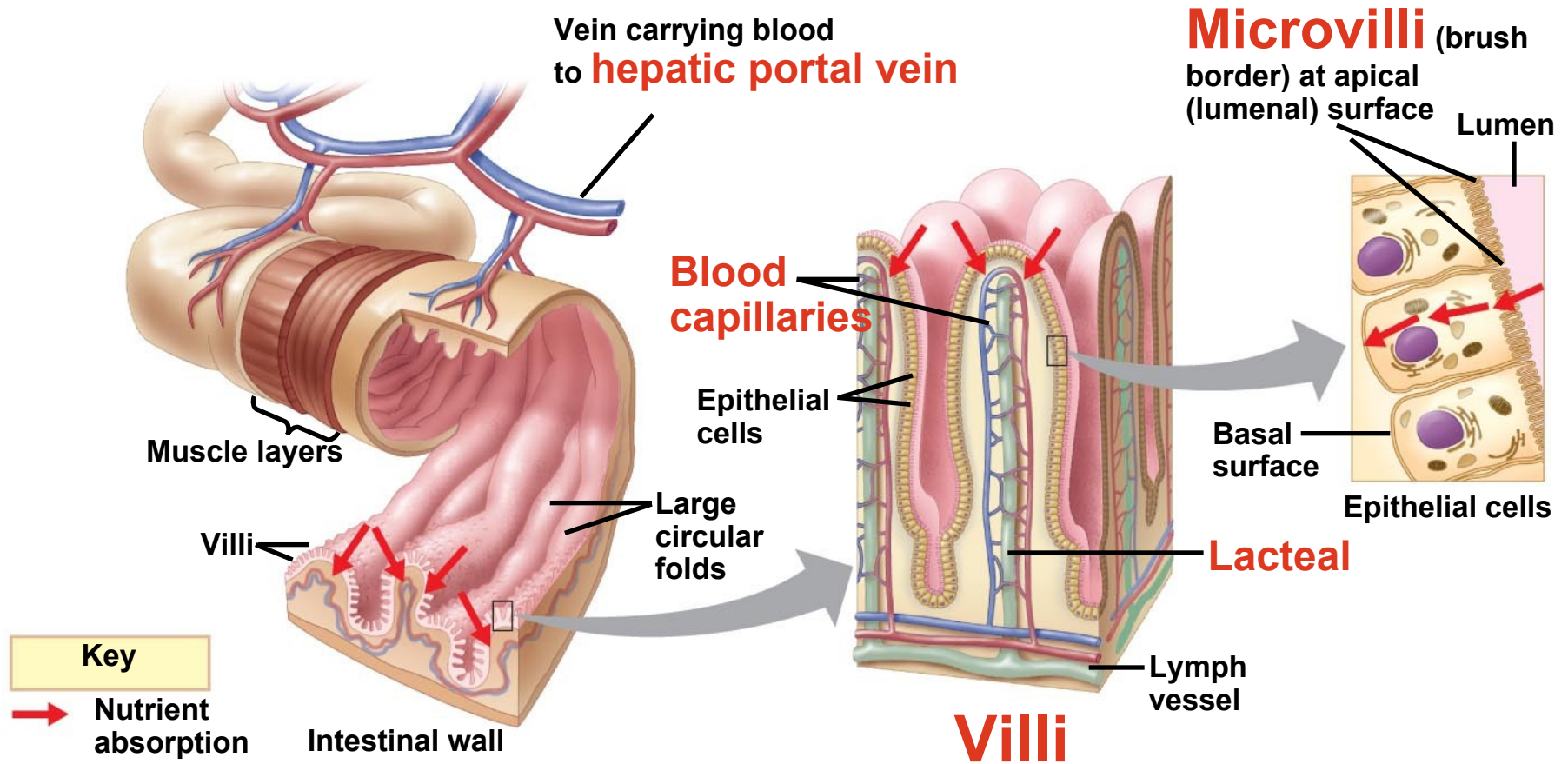
# Absorption in the Small Intestine - Villi

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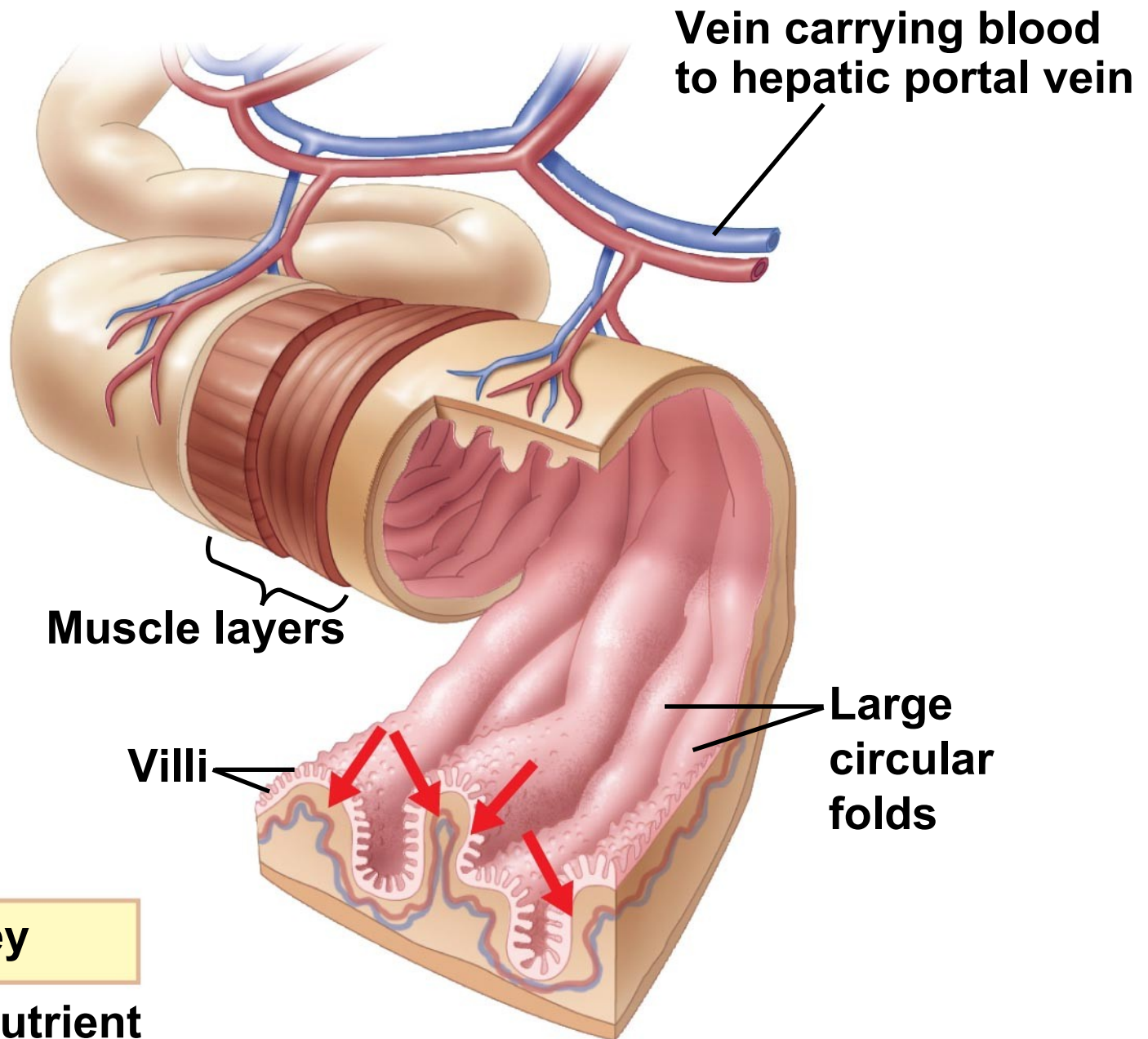
- The *small intestine* has **villi** and **microvilli** that *increase the surface area for absorption*. Villi and microvilli are exposed to the intestinal lumen = space / cavity.
- The enormous microvillar surface area greatly increases the rate of **nutrient absorption**.



# Structure of the small intestine



# Small Intestine

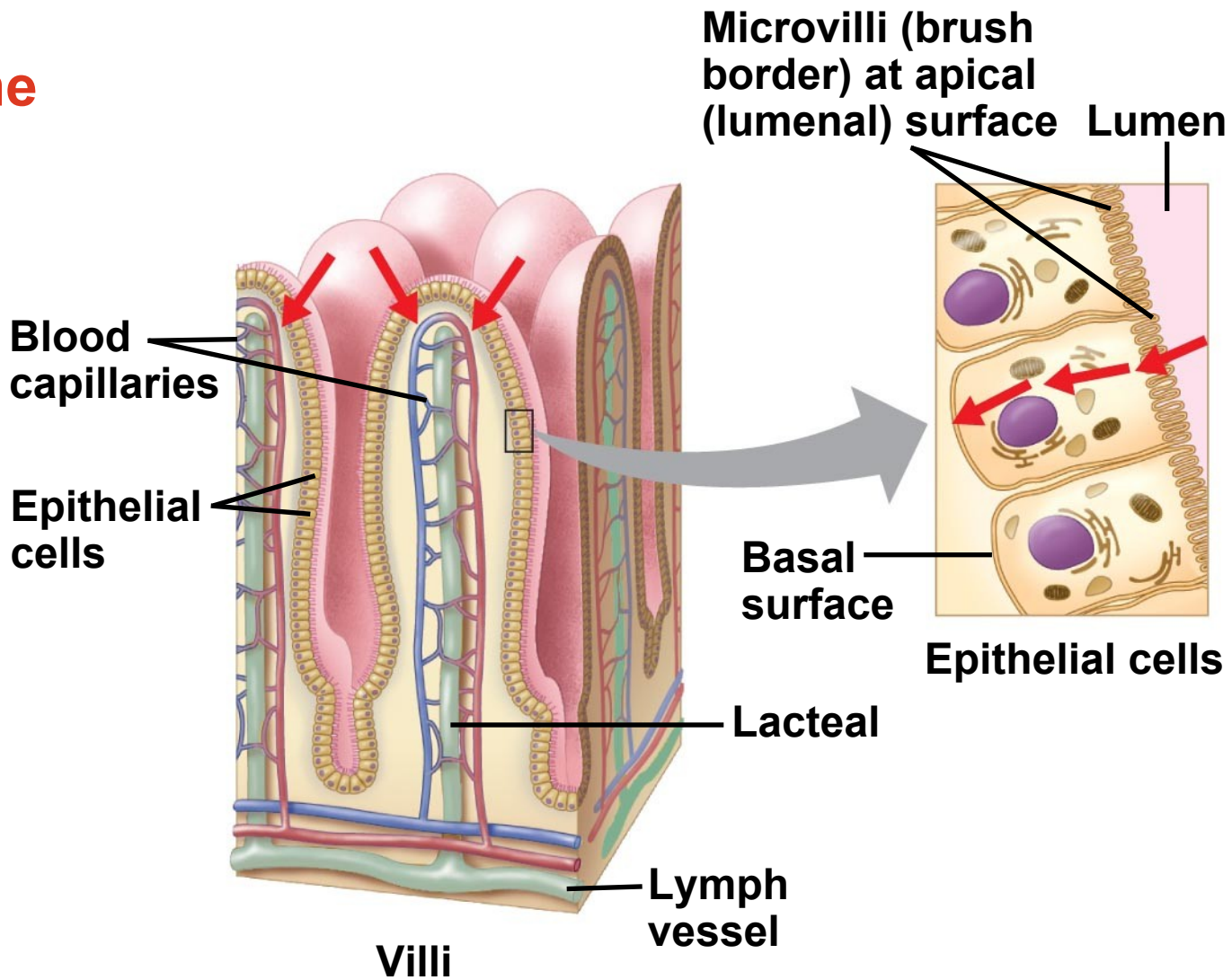


## Key



Nutrient  
absorption

# Small Intestine



## Key



**Nutrient  
absorption**

- 
- Each villus contains a network of blood vessels and a small **lymphatic vessel** called a **lacteal**.
  - After **glycerol** and **fatty acids are absorbed** by epithelial cells, they are recombined into fats within these cells.
  - These *fats are mixed with cholesterol and coated with protein*, forming molecules called **chylomicrons**, which are transported into lacteals.

- 
- **Amino acids and sugars** pass through the epithelium of the small intestine and **enter** the **bloodstream**.
  - Capillaries and veins from the lacteals converge in the **hepatic portal vein** and deliver blood **to the liver** and then on to the heart.

# Absorption in the Large Intestine

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- The **colon** of the **large intestine** is connected to the small intestine.
- The **cecum** aids in the fermentation of plant material and connects where the small and large intestines meet.
- The **human cecum** has an extension called the **appendix**, which plays a very minor role in immunity.



# Digital image of a human colon





- 
- A major function of the **colon** is **water reabsorption**, recovering water that has entered the alimentary canal.
  - **Wastes** of the digestive tract, the **feces**, become more solid as they move through the colon
  - Feces pass through the rectum and exit via the anus.

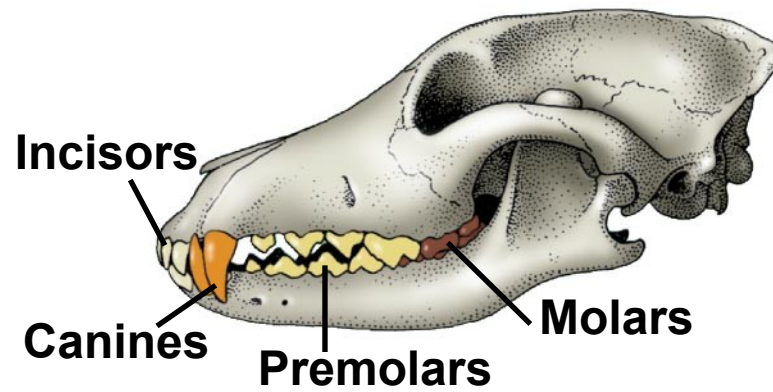
- 
- The L.I. **colon** houses strains of the **bacterium** *Escherichia coli*, some of which **produce** **vitamins ++**.
  - Feces are stored in the **rectum** until they can be eliminated.
  - Two sphincters between the rectum and anus control bowel movements.

# Evolutionary **adaptations** of vertebrate digestive systems correlate with **diet**

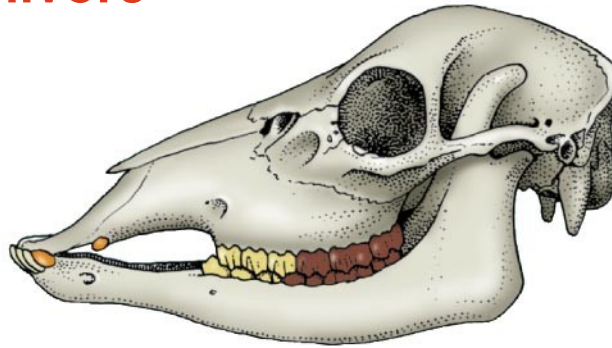
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- Digestive systems of vertebrates are variations on a common plan. There are intriguing adaptations, often related to diet.
- **Dentition**, an animal's assortment of teeth, is one example of structural variation reflecting diet. Mammals have varying dentition that is adapted to their usual diet.
- The **teeth** of poisonous snakes are modified as fangs for injecting **venom**. All snakes can unhinge their **jaws** to swallow prey whole.

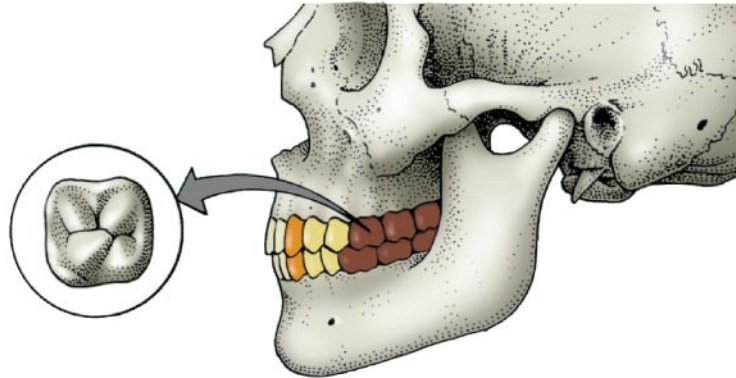
# Dentition and diet



(a) **Carnivore**



(b) **Herbivore**



(c) **Omnivore**

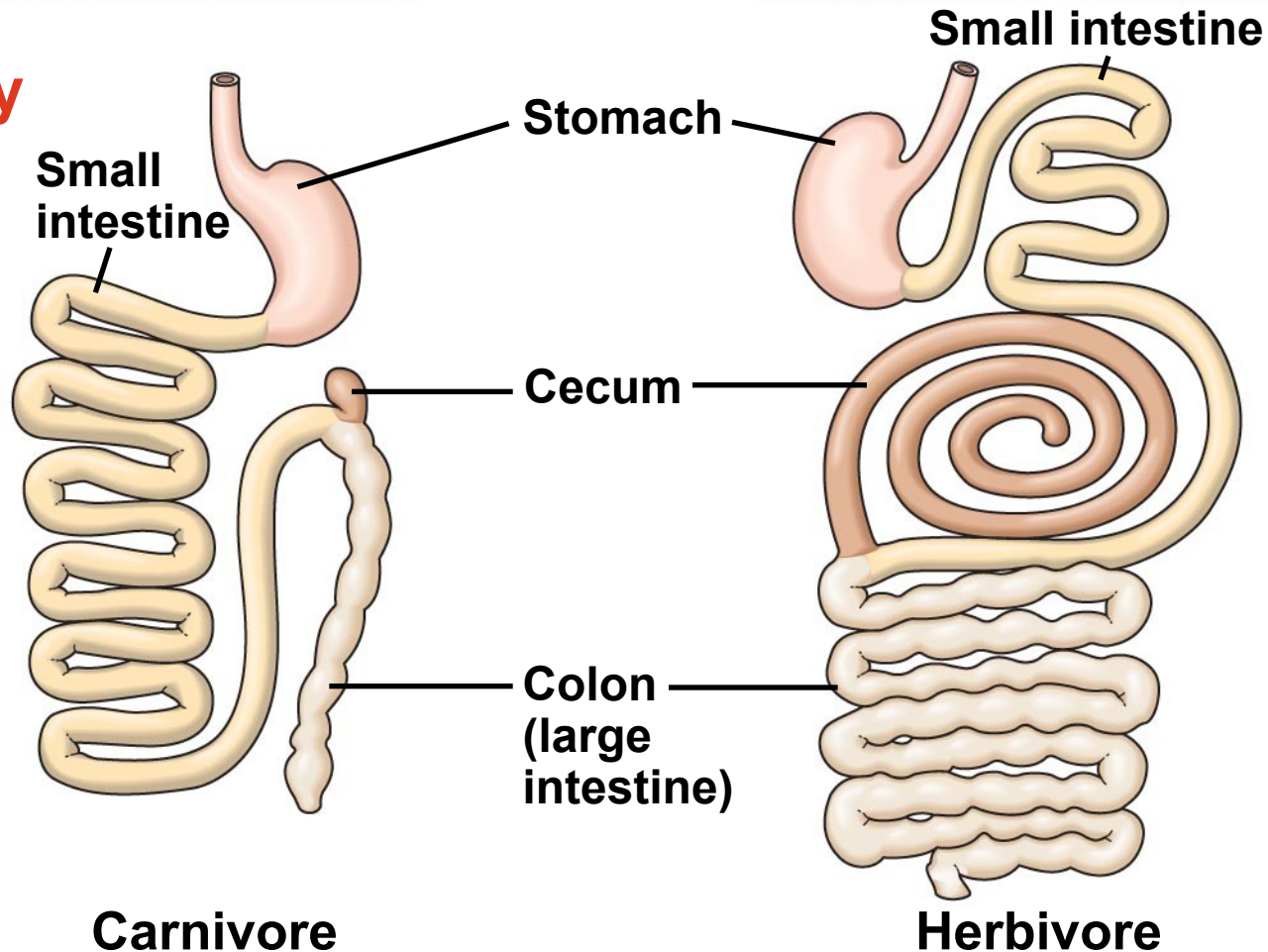
# **Stomach and Intestinal Adaptations**

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- Herbivores generally have longer alimentary canals than carnivores, reflecting the longer time needed to digest vegetation.



**Alimentary  
canals  
of a  
carnivore  
(coyote)  
and  
herbivore  
(koala)**



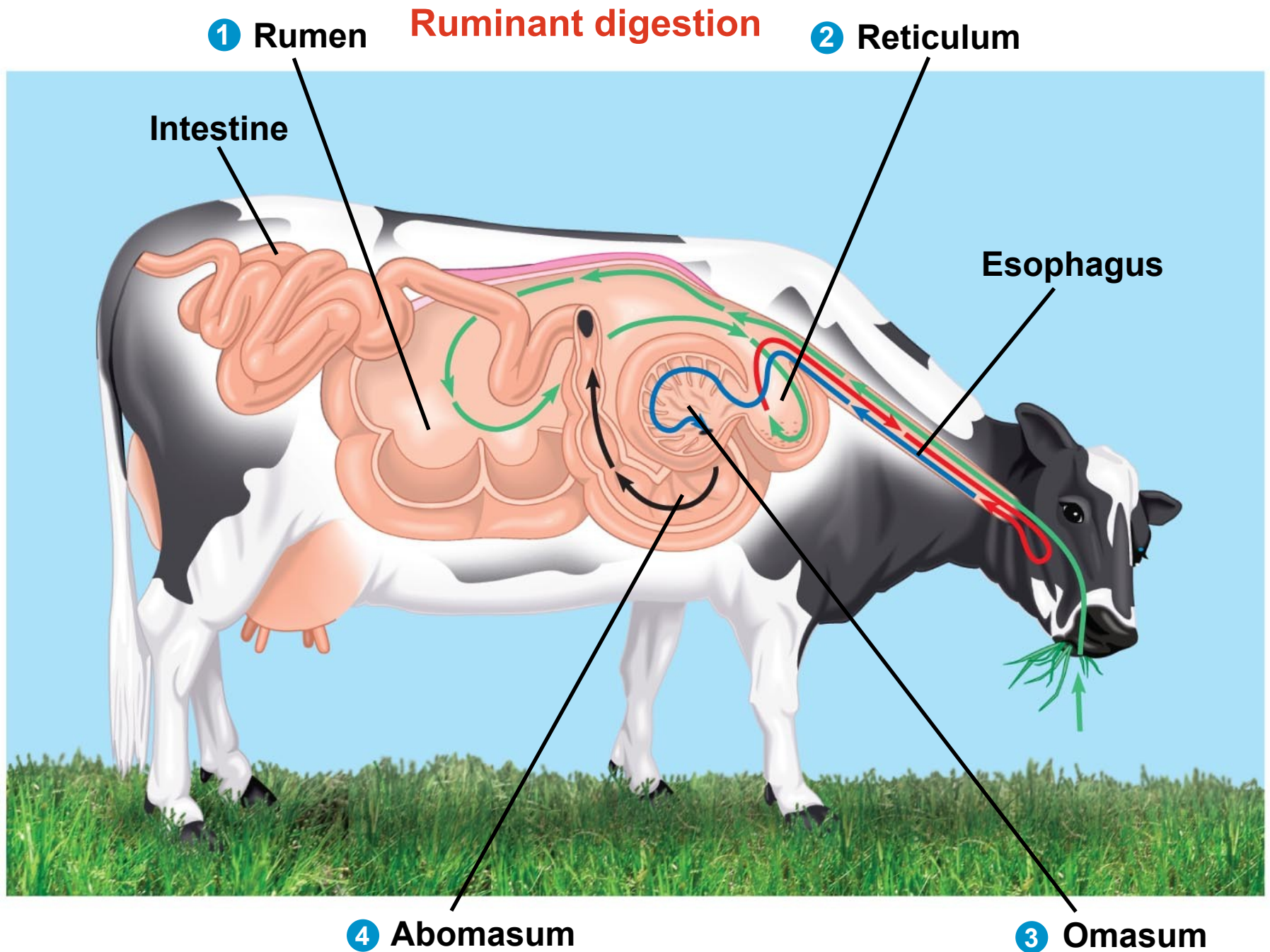


# Mutualistic ++ Adaptations

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- Many herbivores have *fermentation chambers*, where *symbiotic microorganisms ++ digest cellulose*.
- The most elaborate adaptations for an *herbivorous diet* have evolved in the animals called *ruminants*.

# Ruminant digestion

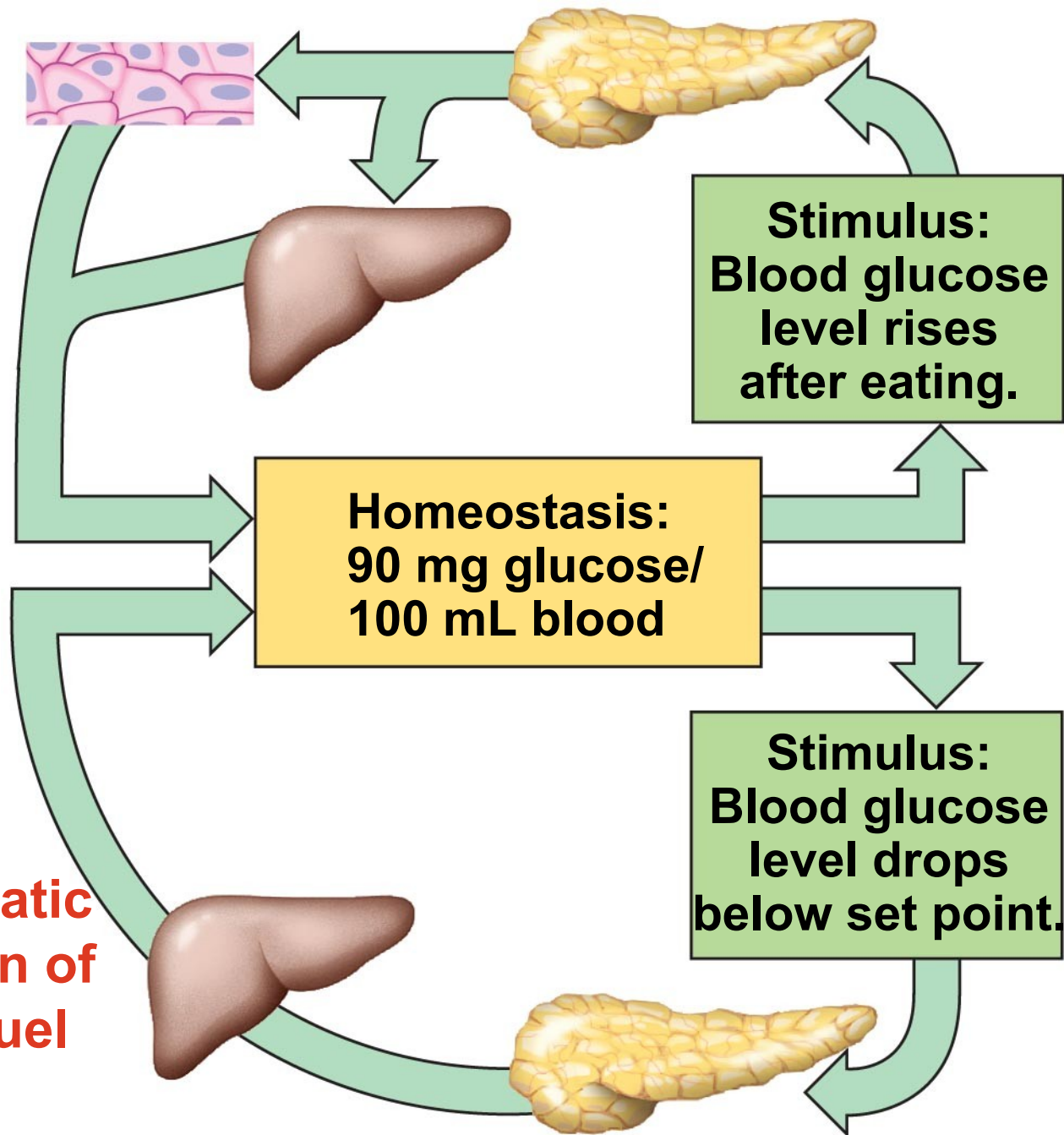


# Homeostatic mechanisms contribute to an animal's energy balance

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- Food energy balances the energy from metabolism, activity, and storage.
- Nearly all of an animal's ATP generation is based on oxidation of energy-rich molecules: carbohydrates, proteins, and fats.
- ***Animals store excess calories primarily as glycogen in the liver and muscles.***
- Energy is secondarily stored as adipose, or fat, cells.
- When fewer calories are taken in than are expended, fuel is taken from storage and oxidized.

# Homeostatic regulation of cellular fuel



# Overnourishment and Obesity

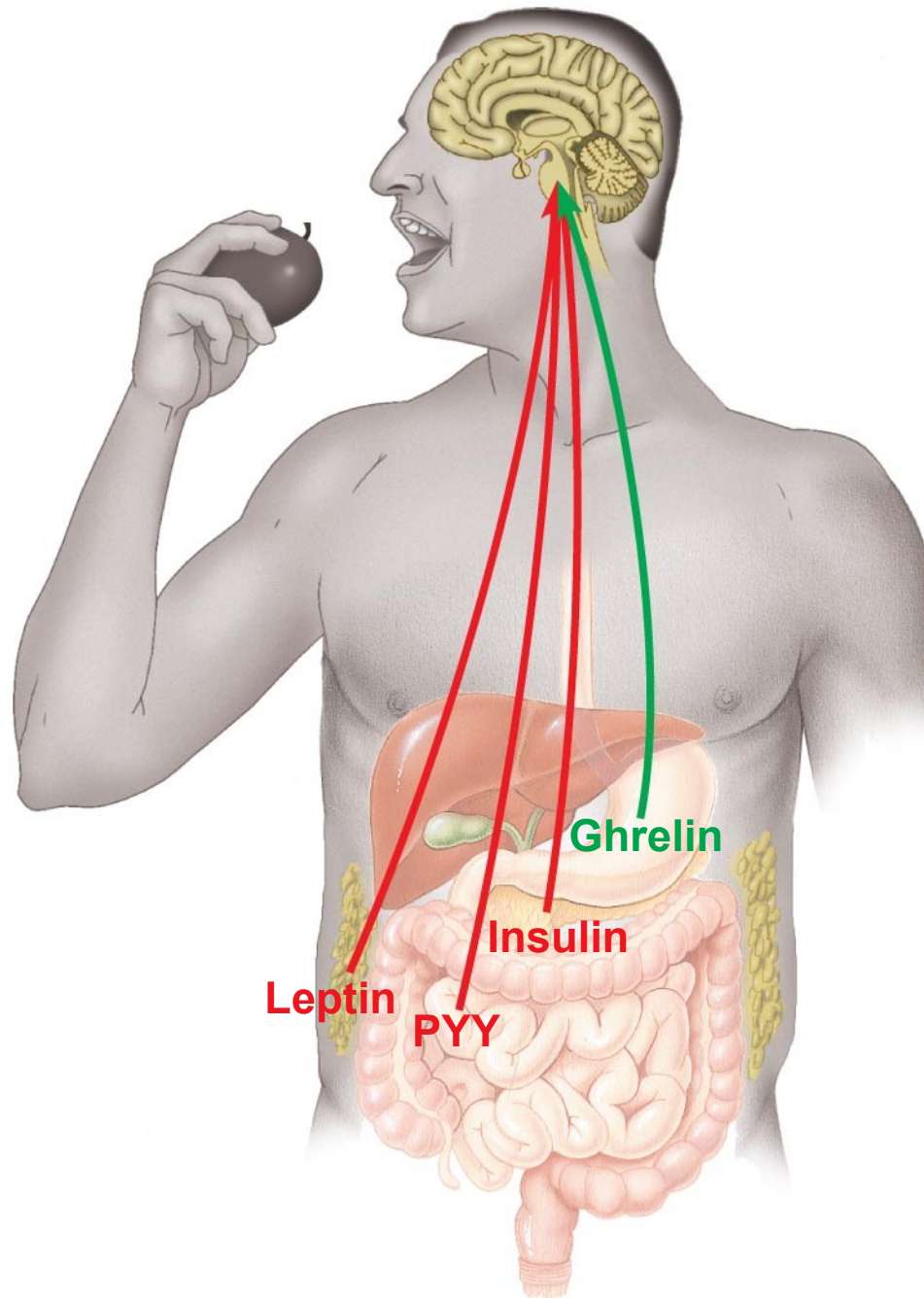
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- **Overnourishment** causes obesity, which results from excessive intake of food energy with the excess stored as fat.
- Obesity contributes to diabetes (type 2), cancer of the colon and breasts, heart attacks, and strokes.
- The complexity of weight control in humans is evident from studies of the hormone leptin.
- Mice that inherit a defect in the gene for an *appetite regulation hormone, leptin*, become very obese.

- 
- Researchers have discovered several of the mechanisms that help regulate body weight.
  - *Homeostatic mechanisms are feedback circuits that control the body's storage and metabolism of fat over the long-term.*
  - Hormones regulate long-term and short-term appetite by affecting a “satiety center” in the brain.



# appetite- regulating hormones





## EXPERIMENT



**Obese mouse with mutant *ob* gene (left) next to wild-type mouse.**

## RESULTS

| Genotype pairing<br>(red type indicates<br>mutant genes; bar<br>indicates pairing) | Average body mass (g) |        |
|------------------------------------------------------------------------------------|-----------------------|--------|
|                                                                                    | Starting              | Ending |
| <i>ob</i> <sup>+</sup> , <i>db</i> <sup>+</sup>                                    | 20.3                  | 23.6   |
| <i>ob</i> <sup>+</sup> , <i>db</i> <sup>+</sup>                                    | 20.8                  | 21.4   |
| <i>ob</i> , <i>db</i> <sup>+</sup>                                                 | 27.6                  | 47.0   |
| <i>ob</i> , <i>db</i> <sup>+</sup>                                                 | 26.6                  | 44.0   |
| <i>ob</i> , <i>db</i> <sup>+</sup>                                                 | 29.4                  | 39.8   |
| <i>ob</i> <sup>+</sup> , <i>db</i> <sup>+</sup>                                    | 22.5                  | 25.5   |
| <i>ob</i> , <i>db</i> <sup>+</sup>                                                 | 33.7                  | 18.8   |
| <i>ob</i> <sup>+</sup> , <i>db</i>                                                 | 30.3                  | 33.2   |

# Obesity and Evolution

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- The problem of maintaining weight partly stems from our evolutionary past, when fat hoarding was a means of survival.
- A species of birds called petrels become obese as chicks; in order to consume enough protein from high-fat food, chicks need to consume more calories than they burn.

# A plump petrel chick





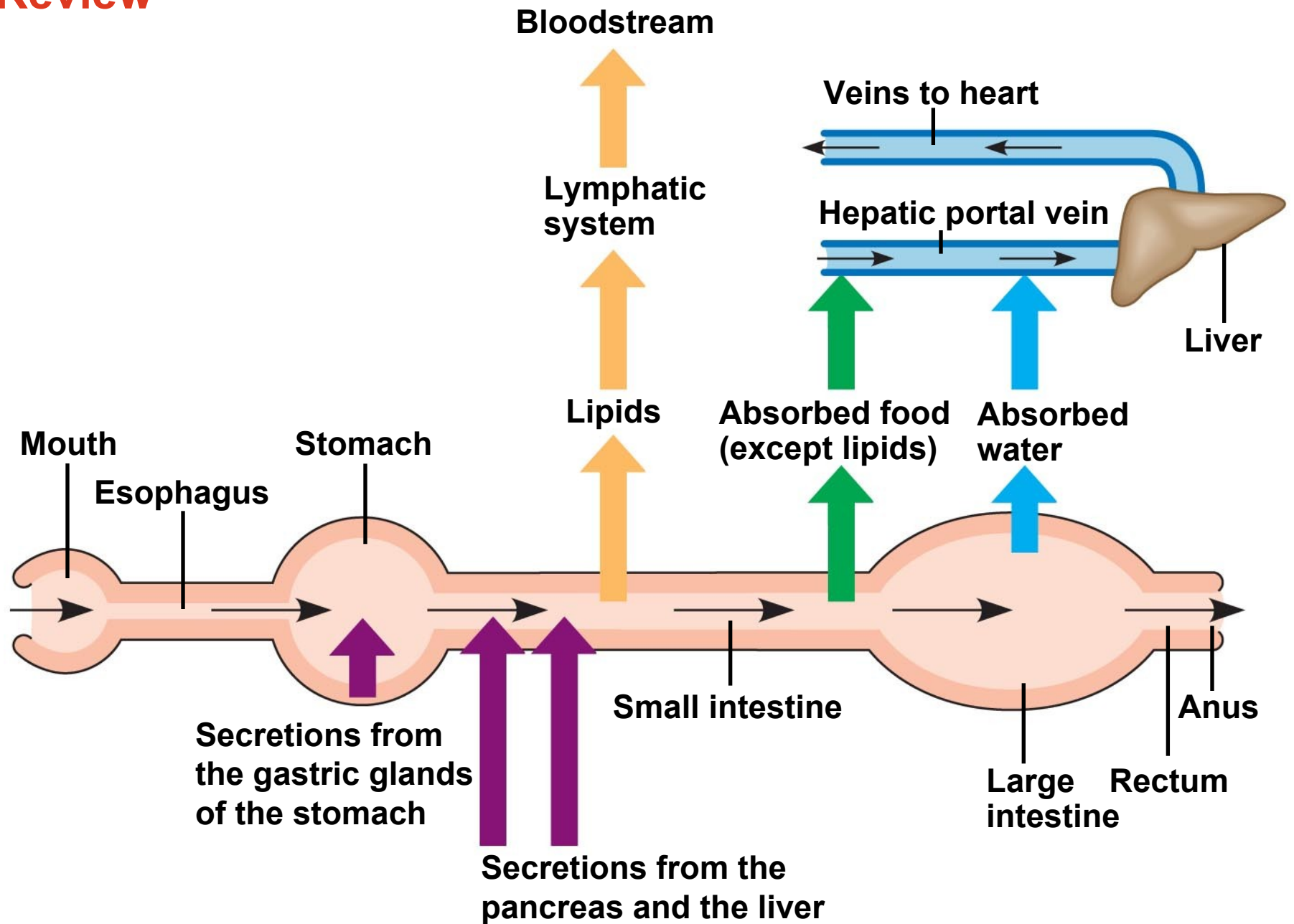
# *Fat cells from the abdomen of a human*

100  $\mu\text{m}$



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# Review



## You should now be able to:

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1. Name the three nutritional needs that must be met by an animal's diet.
2. Describe the four classes of essential nutrients.
3. Distinguish among undernourishment, overnourishment, and malnourishment.
4. Describe the four main stages of food processing.
5. Distinguish between a complete digestive tract and a gastrovascular cavity.

- 
6. Follow a meal through the mammalian digestive system:
    - List important enzymes and describe their roles
    - Compare *where* and *how* the major types of macromolecules are digested and absorbed
  7. Relate variations in dentition with different diets.
  8. Explain *where* and *in what form* energy-rich molecules may be stored in the human body.